

Sharp Euler Bounds, Golden Ladders, and Double Radial Turning Points

Exact certificates and constructive continuation
of the ProveIt polylogarithm manuscript

ProveIt research continuation
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Abstract

We continue the polylogarithm manuscript at the pinned revision specified in Section 1. The first main result settles its open universal real-order Euler remainder problem. For $F_{a,b}(z) = \sum_{n \geq 2} H_{n-1}^{(b)} z^n / n^a$ and all $a, b > 0$, the Euler error satisfies $0 < E_N - \Im F_{a,b}(i) < (9/8)2^{-N}$ for $N \geq 2$. The optimal constant valid simultaneously for every $N \geq 1$ is exactly $\max_{b>0} \{\beta(b) + 2^{-b}\eta(b)\}$; its unique maximizer belongs to the previously proved Gaussian axis theorem. The new ingredient is a positive probability-kernel representation and three finite polynomial positivity certificates, one of which controls every $N \geq 4$.

We also give explicit functional certificates for the two remaining golden weight-five formulas with indices 20 and 24, together with a finite descent to their lower-weight companions. In the real-order zero geometry, an exact rational interval calculation certifies a simultaneous zero of the quadratic and quartic radial coefficients. Its negative sextic coefficient and nonsingular parameter map yield a region with two small-radius turning points, and a fourth-root radius law on a transverse slice. A constructive Cayley quotient projector supplies a further formal obstruction for the frozen S_6 target. The distinction between analytic identities, exact finite certificates, formal relation-space obstructions, and numerical diagnostics is retained throughout. The package includes source, executable verifiers, exact records, proposed manuscript updates, and a focused research agenda.

Scope. The claims of progress are relative to the audited repository and identified previous continuations. Classical identities, the Euler transformation, and the underlying Hopf-algebra machinery are credited as such. No claim of proof-assistant formalization or exhaustive priority assessment is made.

Contents

1	The audited problem and the result boundary	3
1.1	What was already known	3
1.2	Conventions	3
1.3	What a certificate proves	4
2	The optimal universal real-order Euler bound	4
2.1	A probability representation with no threshold	5
2.2	Three polynomial inequalities control every later tail	7
2.3	The finite positivity certificates	8
2.4	Why the first step determines the exact optimum	9
2.5	Consequences for certified computation	10
3	Exact certificates for the remaining golden weight-five ladders	10
3.1	The two evaluations and their full descent	10
3.2	A real generalized Rogers function	11
3.3	A tensor certificate with an elementary analytic interpretation	12
3.4	The two finite certificates	13
3.5	Golden specialization and the endpoint constants	15
3.6	Removing the lower polylogarithms	16
3.7	The first ladder and canonical normalization	17
3.8	Verification, integration, and further questions	18
4	A certified double radial turning-point bifurcation	19
4.1	An exact finite recurrence to every radial order	19
4.2	The sextic coefficient	20
4.3	A unique certified transition in a rational rectangle	21
4.4	The local fold and the region with two extrema	23
4.5	The sextic curve and two radius scaling laws	25
4.6	What the result resolves and what remains	26
5	A constructive normal form for the full Cayley shuffle ideal	26
5.1	The algebra and its precise relation ideal	27
5.2	Endpoint projection that commutes with Cayley symmetry	27
5.3	Explicit eigenvector lifts and the normal-form theorem	28
5.4	An exact obstruction for the frozen weight-seven target	29
5.5	Verification and the next reduction step	31
6	Manuscript updates and a further research program	32
6.1	Corrections of status and scope	32
6.2	Research questions on uniform bounds and asymptotics	33
6.3	Research questions on radial geometry	34
6.4	Research questions on identities and relation spaces	34
6.5	Reproduction and the boundary of the deliverable	35
	References	36

1 The audited problem and the result boundary

The source of this continuation is the collective manuscript *Polylogarithms and their Arithmetic Bridges*, in `Analysis/Polylogarithms/docs/manuscript` of ProveIt [1]. We fixed the repository revision before beginning the mathematical audit. The revision is

3447bc59b78d138a7f0d536b66ac6e5cc5d5e49f.

The reading artifact at that revision has twelve chapters and 375 pages. The audit also inspected the associated research reports and the incoming polylogarithm archives when determining which results had already been obtained. The present deliverable is a separate article for integration; it does not alter the repository remotely.

1.1 What was already known

Several attractive-looking targets were no longer open. In particular, the S_4 evaluation, the stated Gaussian relation-space rank formulas, the signed-measure threshold, principal slit-plane nonvanishing, angular uniqueness, subcritical radial motion, fixed-total Gaussian monotonicity, and the sharp critical Euler constant had already acquired proofs. A previous continuation had also proved the first golden weight-five identity. These facts determine the novelty boundary of the present work: re-establishing one of them is useful only as a clearly credited prerequisite or independent derivation.

The canonical manuscript still explicitly asks for a universal Euler constant outside the critical triangle. It also gives a numerical location at which the quartic radial coefficient vanishes along the quadratic threshold, while leaving the higher local geometry open. The two golden formulas with indices 20 and 24 are printed as numerical identities without executable analytic certificates. The weight-seven S_6 relation remains conjectural. These are the four interfaces addressed here.

Target	Baseline status	Contribution here
Universal Euler error	Parameter-dependent supercritical bounds; sharp critical constant	A universal $9/8$ bound for $N \geq 2$; the exact best constant for all $N \geq 1$
Golden index 20 and 24	Numerical weight-five formulas	Exact functional certificates and a proved finite descent
Quartic transition	Existence and a diagnostic location	Exact local uniqueness, negative sextic term, two turning points and scaling laws
Cayley relations for S_6	Abstract quotient and bounded search obstructions	Explicit quotient projector and a nonzero full-ideal normal form; the special-value identity remains open

1.2 Conventions

For real $a, b > 0$, put

$$H_m^{(b)} = \sum_{j=1}^m j^{-b}, \quad H_0^{(b)} = 0, \quad F_{a,b}(z) = \sum_{n=2}^{\infty} \frac{H_{n-1}^{(b)}}{n^a} z^n. \quad (1)$$

The last series initially defines a holomorphic function on $|z| < 1$. For integer orders it is the depth-two polylogarithm $\text{Li}_{a,b}(z, 1)$ in the nested-sum convention $\sum_{n>m \geq 1} z^n / (n^a m^b)$. Values at i mean the principal analytic continuation, equivalently the radial Abel value. This convention

matters when $a + b \leq 1$, because the ordinary alternating coefficient series need not converge. The positive double integral below defines the continuation without applying a convergence theorem outside its domain.

We use $\beta(s) = \sum_{j \geq 0} (-1)^j / (2j+1)^s$ and $\eta(s) = \sum_{j \geq 1} (-1)^{j-1} / j^s$ for $s > 0$. Both series converge on that domain. In the radial geometry, the conventional symbol $\eta_{a,b}(\rho)$ denotes a normalized angle; its subscripts and argument distinguish it from the Dirichlet eta function. To avoid a collision between the Euler limit and a radial coefficient, the quartic coefficient is always written with its arguments or in a locally specified coordinate system.

The golden ratio and its reciprocal are

$$\phi = \frac{1 + \sqrt{5}}{2}, \quad \rho = \phi^{-1}, \quad \ell = \log \rho = -\log \phi.$$

The golden section uses this constant ρ ; the radial section uses ρ as a variable radius and restates that convention locally. All logarithms of positive real numbers are real logarithms. Real parts of analytically continued polylogarithms at real arguments are specified when intermediate functional identities leave $(0, 1)$.

1.3 What a certificate proves

The Euler positivity verifier performs rational polynomial arithmetic and proves positivity by Bernstein coefficients on a finite partition. The radial verifier evaluates finitely many explicit elementary expressions by outward rational interval arithmetic. The golden certificates are exact tensor identities coupled to a written differential and endpoint argument. The Cayley computation takes place in a specified formal shuffle algebra. Its nonzero normal form proves nonmembership in that algebraic relation ideal, and does not imply nonvanishing of a numerical period.

These are ordinary mathematical proofs with finite computational premises. The executable code and exact records expose those premises for review. Numerical quadrature, plots, and stationary-point solvers serve only as diagnostics. They are neither the source of an exact sign claim nor a substitute for the analytic arguments.

2 The optimal universal real-order Euler bound

Let

$$A_k = \frac{H_{2k}^{(b)}}{(2k+1)^a}, \quad d_j = \sum_{k=0}^j (-1)^k \binom{j}{k} A_k, \quad E_N = \sum_{j=0}^{N-1} \frac{d_j}{2^{j+1}}, \quad g_{a,b} = \Im F_{a,b}(i). \quad (2)$$

Thus $A_0 = d_0 = E_1 = 0$. The Euler transformation itself is classical [2]. The issue addressed here is a parameter-uniform bound with the correct analytic interpretation, including parameter values for which the untransformed alternating series diverges.

Theorem 2.1 (Sharp universal Euler constant). *For all $a, b > 0$ the Euler series is absolutely convergent and E_N decreases strictly to $g_{a,b}$ for $N \geq 1$. Moreover,*

$$0 < E_N - g_{a,b} < \frac{9}{8} 2^{-N} \quad (N \geq 2). \quad (3)$$

Define

$$C_* = \max_{b>0} C(b), \quad C(b) = \beta(b) + 2^{-b} \eta(b). \quad (4)$$

Then

$$0 < E_N - g_{a,b} < C_* 2^{-N} \quad (a, b > 0, N \geq 1), \quad (5)$$

and no smaller constant works simultaneously for all these parameters. The maximum in (4) is attained at exactly one $b = b_* \in (1, 2)$, as in the Gaussian axis theorem of the baseline. On the closure $a \geq 0, b > 0$, equality in the corresponding non-strict bound occurs only at $a = 0, b = b_*, N = 1$.

Numerically,

$$b_* \approx 1.30221658710124120959, \quad C_* \approx 1.13656110333950956095.$$

These decimals are diagnostics, not the definition of the constant. The rigorous result is the exact maximum in (4), its unique maximizer, and the rational bound (3). The latter is particularly convenient for certification: practical evaluations use $N \geq 2$, so they need no evaluation of C_* .

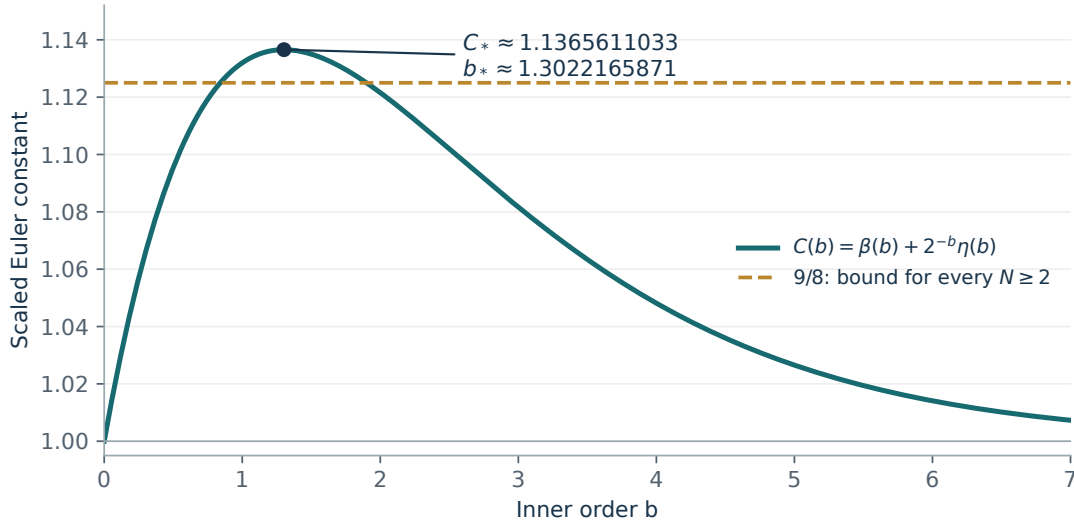


Figure 1: The inherited axis function and the new strict separator for every later Euler step. The curve is a numerical illustration; the maximum is defined exactly by (4), and the separation from $9/8$ has the rational witness (24).

2.1 A probability representation with no threshold

For $s > 0$ let μ_s be the probability measure on $(0, 1)$ with density

$$d\mu_s(x) = \frac{(-\log x)^{s-1}}{\Gamma(s)} dx. \quad (6)$$

Equivalently, $X_s = e^{-T_s}$ where T_s is gamma distributed with shape s and rate one. The elementary gamma integral gives

$$\int_0^1 x^{n-1} d\mu_s(x) = n^{-s} \quad (n > 0).$$

At $s = 0$, when needed, set $\mu_0 = \delta_1$.

Lemma 2.2 (Positive double integral). *For independent $X \sim \mu_a$ and $Y \sim \mu_b$,*

$$A_k = \mathbb{E} \left[X^{2k} \frac{1 - Y^{2k}}{1 - Y} \right], \quad (7)$$

$$F_{a,b}(z) = z^2 \mathbb{E} \left[\frac{X}{(1 - zX)(1 - zXY)} \right] \quad (z \in \mathbb{C} \setminus [1, \infty)). \quad (8)$$

In particular,

$$-g_{a,b} = \mathbb{E} \left[\frac{X^2(1 + Y)}{(1 + X^2)(1 + X^2Y^2)} \right] > 0. \quad (9)$$

These formulas also hold for $a = 0, b > 0$.

Proof. Expand $(1 - Y^m)/(1 - Y) = \sum_{r=0}^{m-1} Y^r$ and apply the gamma moment formula. This proves (7). For $|z| < 1$, absolute convergence permits summation of the two geometric series in

$$\mathbb{E} \left[\frac{z}{1 - Y} \left\{ \frac{1}{1 - zX} - \frac{1}{1 - zXY} \right\} \right].$$

Their difference is exactly the integrand in (8). On a compact subset of the slit plane, both denominators stay uniformly away from zero for $X, Y \in [0, 1]$. The probability integral is therefore holomorphic there and supplies the asserted principal continuation. At $z = i$, rationalization gives (9). No signed-moment existence theorem is required. $\square$

Lemma 2.3 (Exact positive Euler tail). *Set*

$$W_N(x) = \frac{(1 - x^2)^N}{1 + x^2}, \quad \mathcal{K}_N(x, y) = \frac{W_N(xy) - W_N(x)}{1 - y}. \quad (10)$$

Then, for every $N \geq 1$,

$$2^N(E_N - g_{a,b}) = \mathbb{E}[\mathcal{K}_N(X, Y)]. \quad (11)$$

The integrand is strictly positive for $0 < x, y < 1$.

Proof. Apply the binomial theorem to the finite expression in (7):

$$d_j = \mathbb{E} \left[\frac{(1 - X^2)^j - (1 - X^2Y^2)^j}{1 - Y} \right] < 0 \quad (j \geq 1). \quad (12)$$

For $u, v \in [0, 1]$ the inequality $|u^j - v^j| \leq j|u - v|$ gives $|d_j| \leq 2j$. Thus the Euler series converges absolutely, uniformly under the probability integral. Summing it gives

$$\sum_{j \geq 0} 2^{-j-1} d_j = \mathbb{E} \left[\frac{(1 + X^2)^{-1} - (1 + X^2Y^2)^{-1}}{1 - Y} \right] = g_{a,b}.$$

Summing only the terms $j \geq N$ yields (11). Since W_N is strictly decreasing on $(0, 1)$, the sign is strict. The same argument works at $a = 0$, because then $X = 1$ while $0 < Y < 1$. $\square$

The identity (11) separates the uniform problem into an elementary kernel inequality and a probability average. Its first case is

$$\mathcal{K}_1(x, y) = \frac{2x^2(1 + y)}{(1 + x^2)(1 + x^2y^2)}. \quad (13)$$

The behavior of the later kernels is sufficiently different that their bounds cannot be inferred solely from the size of $g_{a,b}$.

2.2 Three polynomial inequalities control every later tail

Lemma 2.4 (Uniform kernel gap). *For every integer $N \geq 2$, $0 \leq x \leq 1$, and $0 \leq y < 1$,*

$$0 \leq \mathcal{K}_N(x, y) < \frac{9}{8}. \quad (14)$$

The same upper bound holds at $y = 1$ by the continuous limiting value.

We give the reduction and all finite premises of the proof. Put $u = x^2$. For $N = 2, 3$, define the polynomial

$$P_N(u, y) = 9(1+u)(1+uy^2) - 8 \frac{(1-uy^2)^N(1+u) - (1-u)^N(1+uy^2)}{1-y}. \quad (15)$$

The numerator vanishes at $y = 1$, so the quotient is a polynomial. Direct division gives

$$P_2 = 9 + u(9y^2 - 24y - 15) + u^2(8y^3 + 17y^2 + 8y + 8) + 8u^3(y^3 + y^2), \quad (16)$$

and

$$P_3 = 9 + u(9y^2 - 32y - 23) + u^2(24y^3 + 33y^2 + 24y + 24) + u^3(-8y^5 - 8y^4 + 16y^3 + 16y^2 - 8y - 8) - 8u^4(y^5 + y^4 + y^3 + y^2). \quad (17)$$

Exact positive Bernstein expansions on dyadic boxes prove $P_2 > 0$ and $P_3 > 0$ on the closed unit square. Since

$$\frac{9}{8} - \mathcal{K}_N(x, y) = \frac{P_N(x^2, y)}{8(1+x^2)(1+x^2y^2)},$$

these prove the first two cases of the kernel lemma.

For $N \geq 4$, write $w_N(t) = (1-t)^N/(1+t)$ for $0 \leq t \leq 1$. Every derivative has the alternating sign appropriate to its order: Leibniz's formula is a sum of products of two factors with that property. Also $w_N^{(j)}(1) = 0$ for $j = 0, 1, 2, 3$. Taylor's formula with integral remainder consequently gives

$$w_N(t) = \int_{(0,1]} (1-t/s)_+^4 d\sigma_N(s), \quad (18)$$

where

$$d\sigma_N(s) = \frac{-s^4 w_N^{(5)}(s)}{24} ds + \frac{w_N^{(4)}(1)}{24} \delta_1(ds). \quad (19)$$

It is a positive measure, and setting $t = 0$ shows that its mass is $w_N(0) = 1$. The atom has mass $1/2$ for $N = 4$ and is absent for $N > 4$. This elementary truncated-power representation is a special case of the classical multiply monotone framework [3, 4]; here Taylor's formula proves exactly the representation used, including its endpoint atom.

For fixed $0 < y < 1$, the largest value over $r \geq 0$ of

$$\frac{(1-r^2y^2)_+^4 - (1-r^2)_+^4}{1-y}$$

occurs at

$$r^2 = \frac{1 - y^{2/3}}{1 - y^{8/3}} < 1.$$

Indeed, in $0 < r < 1$ differentiation in r^2 gives this unique critical point, with derivative positive before it and negative afterwards. For $r \geq 1$ the expression is decreasing until it becomes zero. Its maximum is therefore

$$M_4(y) = \frac{(1 - y)^3(1 + y)^4}{(1 - y^{8/3})^3}. \quad (20)$$

The limiting values at $y = 0, 1$ are also finite. Set $y = v^3$ and define

$$Q_4(v) = \frac{9(1 - v^8)^3 - 8(1 - v^3)^3(1 + v^3)^4}{(1 - v)^3}. \quad (21)$$

The expression is a polynomial of degree 21. Two exact Bernstein expansions, on $[0, 1/2]$ and $[1/2, 1]$, have strictly positive coefficients. Hence $M_4(y) < 9/8$ for $0 \leq y < 1$, and the limiting case also satisfies the strict bound. Substitute (18) into (10), with $r = x/\sqrt{s}$. This ratio may exceed one, which is why the preceding maximization included all $r \geq 0$. A probability average of quantities bounded by $M_4(y)$ proves (14) for all $N \geq 4$.

2.3 The finite positivity certificates

For clarity, the proof certificate does not use a numerical polynomial optimizer. If $p(t) = \sum_{j=0}^d a_j t^j$, its degree- d Bernstein coefficients on $[0, 1]$ are

$$b_k = \sum_{j=0}^k a_j \frac{\binom{k}{j}}{\binom{d}{j}}, \quad p(t) = \sum_{k=0}^d b_k \binom{d}{k} t^k (1 - t)^{d-k}. \quad (22)$$

The basis functions are nonnegative and sum to one. For a box in several variables, first make an affine substitution to the unit box and then apply the formula in each variable. Positivity of every coefficient proves strict positivity on the whole closed box.

The standard-library verifier `code/certify_euler_bound.py` constructs the three polynomials directly from (15) and (21), checking each exact polynomial division. It recursively bisects a box only when its current Bernstein coefficients do not already prove positivity. The resulting finite partition and its exact rational lower coefficients are written to `data/euler_polynomial_certificate.json`.

Polynomial	Boxes	Positive coefficients	Smallest coefficient
P_2	13	208	3/64
P_3	7	210	11337/40960
Q_4	2	44	1

Thus the proof uses 462 positive rational coefficients in total. An independent SymPy verifier reconstructs the expressions without importing the producer's polynomial operations, checks all coefficients, checks that box interiors are disjoint, and checks that their total volume is one. The producer's recursive bisection already ensures coverage; the independent checks also make the stored partition directly reviewable.

2.4 Why the first step determines the exact optimum

For fixed $y \in (0, 1)$, (13) is strictly increasing in $x \in (0, 1)$, because its derivative has the positive factor $1 - x^4 y^2$. Consequently

$$-2g_{a,b} = \mathbb{E}[\mathcal{K}_1(X, Y)] < \mathbb{E}\left[\frac{1+Y}{1+Y^2}\right] = C(b) \quad (a > 0). \quad (23)$$

The equality on the right follows either from the axis formula $F_{0,b}(z) = z \operatorname{Li}_b(z)/(1-z)$ or by the convergent residue-class definitions of β and η . As $a \downarrow 0$, $X \rightarrow 1$ in probability, so bounded convergence gives equality in the limit.

The following prerequisite was already proved in the baseline. We include its principal argument to make the use of the maximum self-contained.

Proposition 2.5 (The inherited axis maximum). *The function C in (4) has exactly one stationary point on $(0, \infty)$. It is a nondegenerate global maximum. Moreover, $C(b) > 1$ for finite $b > 0$, and $C(b) \rightarrow 1$ at both ends of the interval.*

Proof. Let $\varphi(T) = (1 + e^{-T})/(1 + e^{-2T})$. Then $C(b) = \mathbb{E}[\varphi(T_b)]$. The function φ increases and then decreases, its unique maximum being at $T = \log(1 + \sqrt{2})$. It is larger than one and at most $(1 + \sqrt{2})/2$ for finite $T > 0$. Gamma concentration at zero as $b \downarrow 0$, and at infinity as $b \rightarrow \infty$, gives the endpoint limits and existence of a maximum.

For $1 < \lambda < \max \varphi$, the function $f(T) = \varphi(T) - \lambda$ has exactly two sign changes $T_1 < T_2$, with signs negative, positive, negative. It is impossible that $C(b_j) = \lambda$ at three distinct $b_1 < b_2 < b_3$. To see this, put $v = T^{b_2-b_1}$ and $p = (b_3 - b_1)/(b_2 - b_1) > 1$. Subtract from v^p the secant line through its values at T_1, T_2 , and multiply by $-T^{b_1-1}$. Strict convexity produces a linear combination of the three powers T^{b_j-1} with the same strict sign pattern as f . Its integral against $f(T)e^{-T}$ is positive, contradicting the three vanishing Mellin integrals.

At a stationary point b_0 , put $\lambda = C(b_0)$. If $C''(b_0) = 0$ as well, the Mellin integral of f and its first two derivatives in b vanish. Their linear combination with kernel

$$-T^{b_0-1}(\log T - \log T_1)(\log T - \log T_2)$$

again gives a strictly positive integral, a contradiction. Every stationary point is therefore nondegenerate. A local minimum would give three intersections with its horizontal level, since the two endpoint limits are lower; hence every stationary point is a strict maximum. There can be only one. All differentiations are justified by the exponential tail and integrable powers of $|\log T|$ at zero. $\square$

The finer bracket $1 < b_* < 2$ is inherited from the baseline's exact comparison at $b = 3/2$ and its elementary endpoint bounds. The new Euler theorem only needs uniqueness and $C_* > 9/8$. Even that last strict inequality has a very short independent rational witness:

$$C_* \geq C(1) > -2E_8(0, 1) = \frac{1627631}{1441440} > \frac{9}{8}. \quad (24)$$

The finite value uses rational harmonic numbers and (2); the strict first comparison follows from the negative Euler increments. It is checked by the same exact script.

Proof of Theorem 2.1. Absolute convergence, strict decrease, and the exact tail follow from Lemma 2.3. The kernel gap gives (3). For $N = 1$, $E_1 = 0$, and (23) bounds the scaled error by C_* . For every $N \geq 2$, (24) makes the universal $9/8$ bound strictly smaller than C_* . This proves (5).

Fix $b = b_*$ and let $a \downarrow 0$. At $N = 1$ the scaled error converges to C_* . No smaller universal constant is possible. For $a = 0$ equality is attained at $b = b_*$, $N = 1$; uniqueness of the axis maximum, strictness in (23), and the gap for $N \geq 2$ exclude every other equality case. $\square$

2.5 Consequences for certified computation

The new bound removes the parameter restrictions from the error budget. Given an exact or outward-enclosed finite Euler sum with $N \geq 2$, the analytic Gaussian value lies in

$$E_N - \frac{9}{8 \cdot 2^N} < g_{a,b} < E_N. \quad (25)$$

If the finite sum itself is enclosed by $[L_N, U_N]$, use $[L_N - 9/(8 \cdot 2^N), U_N]$. Fractional-power rounding errors must be included in $[L_N, U_N]$; the theorem certifies truncation error only. For a requested truncation tolerance $\varepsilon > 0$, any

$$N \geq \max\left(2, \left\lceil \log_2 \frac{9}{8\varepsilon} \right\rceil\right)$$

suffices. No signed-kernel mass, parameter-dependent zeta bound, or numerical estimate of b_* enters this choice.

The finite binomial-tail formula from the baseline remains useful:

$$E_N = 2^{-N} \sum_{k=0}^{N-1} (-1)^k A_k \sum_{j=k+1}^N \binom{N}{j}. \quad (26)$$

It follows by interchanging the two finite sums in (2) and applying Pascal's identity. Its coefficients allow exact rational accumulation when the orders are integers, and interval accumulation otherwise.

Remark 2.6 (What remains unoptimized). The number $9/8$ is a convenient certified separator, not a claim of the least constant restricted to $N \geq 2$. Numerically the pointwise kernel maximum at $N = 2$ is about 1.11894, below $9/8$; averaging against the gamma laws may lower the optimum further. Neither observation is used in the proof. The all- N optimum is nevertheless exact because the first-step supremum is strictly larger than the proved bound on every later step.

3 Exact certificates for the remaining golden weight-five ladders

Put

$$\rho = \frac{\sqrt{5} - 1}{2}, \quad \phi = \rho^{-1}, \quad \lambda = \log \rho = -\log \phi.$$

The golden-ratio discussion in the audited manuscript identifies three weight-five evaluations with the index-12, 20, and 24 ladder families, and records numerical evidence for them [1]. The first evaluation has already received an analytic proof in the preceding research. The contribution here is a complete local proof certificate for each of the other two evaluations, together with all its lower-weight companions. These are statements about what is now proved explicitly in the repository's research program; they do not assert historical priority for the ladder evaluations. The general connection between rational-function identities, tensor kernels, and polylogarithm ladders is classical; see Gangl [6].

The certificates use rational functions of one variable. Consequently, the lower-weight conclusions follow from identities holding on an open interval. No differentiation of an identity valid only at the isolated algebraic number ρ is involved.

3.1 The two evaluations and their full descent

Define two finitely supported coefficient vectors by

$$(c_2^{\text{II}}, c_4^{\text{II}}, c_{10}^{\text{II}}, c_{20}^{\text{II}}) = (45000, -16875, -144, 9), \quad (27)$$

$$(c_2^{\text{III}}, c_4^{\text{III}}, c_6^{\text{III}}, c_8^{\text{III}}, c_{24}^{\text{III}}) = (15660, -19440, 7680, 2430, -15). \quad (28)$$

A missing coefficient means zero. The superscripts refer to the second and third displayed golden weight-five formulas in the manuscript; the third vector is the displayed formula before the canonical index-24 recombination.

Theorem 3.1 (Golden evaluations with exact descent). *For either vector $c = c^{\text{II}}, c^{\text{III}}$, put*

$$S_m = \sum_{j \geq 1} c_j j^{5-m} \text{Li}_m(\rho^j), \quad 1 \leq m \leq 5.$$

Then

$$S_m = \frac{a\lambda^m}{m!} + \sum_{r=2}^m q_r \zeta(r) \frac{\lambda^{m-r}}{(m-r)!}, \quad (29)$$

where the constants are

	a	q_2	q_3	q_4	q_5
II	720000	-57600	0	32040	28944
III	-250560	8640	0	2520	5025

In particular, writing $\ell = \log \phi$, the two weight-five evaluations are

$$\begin{aligned} & 45000 \text{Li}_5(\rho^2) - 16875 \text{Li}_5(\rho^4) - 144 \text{Li}_5(\rho^{10}) + 9 \text{Li}_5(\rho^{20}) \\ &= 28944 \zeta(5) - 6000 \ell^5 + 1600 \pi^2 \ell^3 - 356 \pi^4 \ell, \end{aligned} \quad (30)$$

$$\begin{aligned} & 15660 \text{Li}_5(\rho^2) - 19440 \text{Li}_5(\rho^4) + 7680 \text{Li}_5(\rho^6) + 2430 \text{Li}_5(\rho^8) - 15 \text{Li}_5(\rho^{24}) \\ &= 5025 \zeta(5) + 2088 \ell^5 - 240 \pi^2 \ell^3 - 28 \pi^4 \ell. \end{aligned} \quad (31)$$

The vanishing $q_3 = 0$ matters: the descent produces no $\zeta(3)\lambda^2$ term at weight five. This cancellation will follow from the exact endpoint arithmetic of the functional certificates.

3.2 A real generalized Rogers function

For $n \geq 2$ and real $x \neq 0, 1$, define

$$\begin{aligned} \mathcal{R}_n(x) &= \sum_{r=0}^{n-2} \frac{(-\log |x|)^r}{r!} \text{Re Li}_{n-r}(x) \\ &\quad + \frac{(-1)^{n-1}(n-1)}{n!} \log^{n-1} |x| \text{Re Li}_1(x). \end{aligned} \quad (32)$$

Here the ordinary polylogarithms use their principal branches, and $\text{Re Li}_1(x) = -\log |1-x|$. We extend the definition continuously by $\mathcal{R}_n(0) = 0$ and $\mathcal{R}_n(1) = \zeta(n)$. The first functions are

$$\begin{aligned} \mathcal{R}_2(x) &= \text{Re Li}_2(x) + \frac{1}{2} \log |x| \log |1-x|, \\ \mathcal{R}_3(x) &= \text{Re Li}_3(x) - \log |x| \text{Re Li}_2(x) - \frac{1}{3} \log^2 |x| \log |1-x|, \\ \mathcal{R}_4(x) &= \text{Re Li}_4(x) - \log |x| \text{Re Li}_3(x) + \frac{1}{2} \log^2 |x| \text{Re Li}_2(x) - \frac{1}{8} \log^3 |x| \text{Re Li}_1(x), \\ \mathcal{R}_5(x) &= \text{Re Li}_5(x) - \log |x| \text{Re Li}_4(x) + \frac{1}{2} \log^2 |x| \text{Re Li}_3(x) \\ &\quad - \frac{1}{6} \log^3 |x| \text{Re Li}_2(x) + \frac{1}{30} \log^4 |x| \text{Re Li}_1(x). \end{aligned}$$

In particular, $\mathcal{R}_3$ is the usual real single-valued trilogarithm. At higher weights it is useful to retain $\mathcal{R}_n$ itself, including at even weights.

Lemma 3.2 (Derivative, inversion, and duplication). *Write $L = \log |x|$ and $M = \log |1 - x|$. On each real interval avoiding 0 and 1,*

$$d\mathcal{R}_n(x) = \frac{(-1)^n(n-1)}{n!} L^{n-2} (L dM - M dL). \quad (33)$$

Let

$$h_n(+1) = 1, \quad h_n(-1) = -(1 - 2^{1-n}).$$

For nonzero real x ,

$$\mathcal{R}_n(x^{-1}) = (-1)^{n-1} \mathcal{R}_n(x) + (1 - (-1)^{n-1}) h_n(\operatorname{sgn} x) \zeta(n), \quad (34)$$

$$\mathcal{R}_n(x) + \mathcal{R}_n(-x) = 2^{1-n} \mathcal{R}_n(x^2). \quad (35)$$

Proof. The derivative relation $d \operatorname{ReLi}_j(x) = \operatorname{ReLi}_{j-1}(x) dL$ holds for $j \geq 2$. Consequently the derivative of

$$F_n(x) = \sum_{r=0}^{n-1} \frac{(-L)^r}{r!} \operatorname{ReLi}_{n-r}(x)$$

telescopes to $(-1)^n L^{n-1} dM / (n-1)!$. Since

$$\mathcal{R}_n = F_n + \frac{(-1)^{n-1}}{n!} L^{n-1} M,$$

one more differentiation proves (33). Under inversion, (L, M) becomes $(-L, M - L)$. Substitution in the derivative formula shows that the difference in (34) is constant separately on each of the positive and negative half-axes. Evaluation at $x = 1$ and $x = -1$ fixes the constants, because

$$\mathcal{R}_n(-1) = \operatorname{Li}_n(-1) = -(1 - 2^{1-n}) \zeta(n).$$

The duplication formula follows term by term from the ordinary polylogarithm duplication formula and $\log(x^2) = 2 \log |x|$. These arguments extend across $x = 1$ by continuity. In (32), the potentially singular monologarithm is multiplied by a positive power of $\log |x|$, and its product tends to zero as $x \rightarrow 1$. $\square$

3.3 A tensor certificate with an elementary analytic interpretation

Let $V = \mathbb{Q}(t)^\times \otimes_{\mathbb{Z}} \mathbb{Q}$, with the multiplicative group written additively. Signs are torsion and therefore vanish in V . For a nonzero rational function $f \neq 1$, put

$$\beta_n[f] = f^{\otimes(n-2)} \otimes (f \wedge (1 - f)).$$

The repeated first factors can equivalently be placed in a symmetric tensor power. Only elementary linear algebra with this notation is needed below.

The rational functions in the certificates have the form

$$f_{s,a,b,c}(t) = s t^a (1 - t)^b (1 + t)^c, \quad s \in \{1, -1\}. \quad (36)$$

They have neither zeros nor poles for $0 < t < 1$. Define

$$k(a, b, c) = a + 2b - c. \quad (37)$$

This integer is the exponent after golden specialization, since $1 - \rho = \rho^2$ and $1 + \rho = \rho^{-1}$.

Lemma 3.3 (Exact descent from a rational-function certificate). *Suppose $f_i = f_{s_i, a_i, b_i, c_i}$ and $u_i \in \mathbb{Q}$ satisfy*

$$\sum_i u_i \beta_5[f_i] = 0. \quad (38)$$

Write $k_i = a_i + 2b_i - c_i$. For $2 \leq n \leq 5$, the function

$$H_n(t) = \sum_i u_i k_i^{5-n} \mathcal{R}_n(f_i(t)) \quad (39)$$

is constant on $0 < t < 1$. If all $a_i \geq 0$, its value is

$$H_n(t) = E_n \zeta(n), \quad E_n = \sum_{a_i=0} u_i k_i^{5-n} h_n(s_i). \quad (40)$$

We use $k_i^0 = 1$, including when $k_i = 0$.

Proof. Extend the assignments

$$\eta(t) = 1, \quad \eta(1-t) = 2, \quad \eta(1+t) = -1$$

to a linear functional $\eta : V \rightarrow \mathbb{Q}$. For example, assign zero to every other primitive irreducible polynomial and every rational prime. Then $\eta(f_i) = k_i$. Contracting $5-n$ of the first three tensor factors in (38) gives

$$\sum_i u_i k_i^{5-n} \beta_n[f_i] = 0. \quad (41)$$

For a fixed regular t , evaluate a tensor factor g by $\log |g(t)|$, and evaluate a wedge $g \wedge h$ by

$$\log |g(t)| d \log |h(t)| - \log |h(t)| d \log |g(t)|.$$

Equation (41), together with (33), gives $dH_n = 0$. There are only finitely many points in $(0, 1)$ where some $f_i(t) = 1$. The functions $\mathcal{R}_n(f_i(t))$ are continuous at these points, so the constants on the intervening intervals agree. Finally, if $a_i > 0$ then $f_i(t) \rightarrow 0$ as $t \rightarrow 0^+$, while if $a_i = 0$ then $f_i(t) \rightarrow s_i$. These limits prove (40). $\square$

A rational constant in $1 - f_i$ must be retained when verifying the tensor equation. For example,

$$1 - \frac{1-t}{1+t} = \frac{2t}{1+t}$$

contributes the independent constant-prime coordinate 2 to $1 - f_i$. It has logarithm $\log 2$ and derivative zero; discarding it would change both the tensor equation and its analytic interpretation. The certificates below and the verification program retain this coordinate.

3.4 The two finite certificates

Table 1 defines coefficients $u_{s,a,b,c}$ and $v_{s,a,b,c}$. An entry 0 means that the corresponding argument is absent. There are 58 nonzero u entries and 60 nonzero v entries. For each column separately,

$$\sum_{s,a,b,c} u_{s,a,b,c} \beta_5[f_{s,a,b,c}] = 0, \quad \sum_{s,a,b,c} v_{s,a,b,c} \beta_5[f_{s,a,b,c}] = 0. \quad (42)$$

Thus the table also gives two explicit rational-function identities (39), and four weights of each identity. Their endpoint constants are displayed below.

Here is a completely finite interpretation of the two checks in (42). Factor every f and $1 - f$ over $\mathbb{Q}(t)$, including the rational contents. Use rational primes and primitive irreducible polynomials as a multiplicative basis modulo signs. Write the factor vectors as e_f and d_f . For each unordered pair of basis indices (p, q) and each triple (i, j, k) chosen with repetition from the three coordinates $t, 1 - t, 1 + t$, verify

$$\sum_f u_f(e_f)_i(e_f)_j(e_f)_k((e_f)_p(d_f)_q - (e_f)_q(d_f)_p) = 0, \quad (43)$$

and then the same equation with u replaced by v . The first column has 16 basis coordinates, including the prime 2; the second has 18, also including 2. There are respectively 330 and 370 potentially nonzero weight-five coordinate sums, and every one is zero. These are exact integer equalities. The supplied verifier factors the functions afresh and performs these sums using rational arithmetic. It additionally checks all four contracted tensor equations directly.

Table 1: The two weight-five rational-function certificates. The argument in a row is $s t^a(1 - t)^b(1 + t)^c$.

s	a	b	c	u	v
1	0	0	1	-2166	-1161
-1	0	0	1	-3240	-2025
-1	0	0	3	40	25
1	0	1	-1	-80	-288
-1	0	1	-1	-80	-288
1	0	1	0	2178	3510
-1	0	1	0	2916	4536
1	0	1	1	6	-270
1	0	1	2	26	72
-1	0	2	-2	10	36
1	0	2	1	-54	-27
-1	0	3	0	-36	-56
-1	1	-3	-2	8	9
1	1	-2	-3	-4	-9
-1	1	-2	-1	-40	0
-1	1	-2	0	284	405
1	1	-1	-1	-6	27
-1	1	-1	-1	-162	-351
1	1	-1	0	2268	2997
-1	1	-1	0	1722	2430
-1	1	-1	1	-105	-360
1	1	0	-2	-320	-144
1	1	0	-1	-1710	-81
-1	1	0	-1	-2592	-486
1	1	0	0	-1236	-54
-1	1	0	0	528	1539
1	1	0	1	-126	-414
-1	1	0	1	-480	-405
1	1	1	-1	-70	-270
1	1	1	0	444	909
-1	1	1	0	-42	9
1	1	1	1	-72	-81
-1	1	1	1	42	81
-1	1	2	1	0	9
-1	1	3	1	5	0

Table 1 continued

s	a	b	c	u	v
-1	2	-2	-1	-72	-81
-1	2	-1	-2	32	81
-1	2	-1	-1	6	-270
1	2	-1	0	-206	-504
-1	2	-1	0	204	234
1	2	0	-1	38	99
-1	2	0	-1	-240	270
-1	2	0	0	-210	-540
1	2	1	0	-60	0
-1	2	1	0	0	81
1	3	-3	-3	0	-1
1	3	-3	0	-28	-37
1	3	-2	-1	-46	-9
-1	3	-1	-2	18	54
-1	3	-1	-1	-30	0
-1	3	-1	0	-72	-81
1	3	-1	1	14	45
-1	3	0	-3	32	6
1	3	0	-1	12	81
-1	3	0	-1	0	81
-1	3	0	0	4	19
1	4	-3	-2	4	0
-1	4	-1	-2	0	9
-1	4	0	1	8	9
-1	4	1	0	-8	-9
1	5	-3	-1	8	9
-1	5	-1	-3	-3	-9
1	5	1	-1	0	9
-1	6	3	0	0	-1
-1	7	1	-1	1	0
-1	9	0	-3	0	-1

The table was obtained by exact linear algebra on the functions (36) with numerator and denominator degrees at most 9, after cancellation of common factors. The search discovers the rows; the explicit table and the coordinate equations (43) are the proof certificate. A complete classification of all functional identities in this or a larger alphabet is not needed for the argument.

3.5 Golden specialization and the endpoint constants

At $t = \rho$, the table argument becomes $s\rho^k$, with $k = a + 2b - c$. Applying inversion to negative exponents and duplication to negative arguments gives

$$\sum_f u_f k_f^{5-n} \mathcal{R}_n(f(\rho)) = \frac{1}{144} \sum_{j \geq 1} c_j^{\text{II}} j^{5-n} \mathcal{R}_n(\rho^j) + B_n^{\text{II}} \zeta(n), \quad (44)$$

$$\sum_f v_f k_f^{5-n} \mathcal{R}_n(f(\rho)) = \frac{1}{120} \sum_{j \geq 1} c_j^{\text{III}} j^{5-n} \mathcal{R}_n(\rho^j) + B_n^{\text{III}} \zeta(n). \quad (45)$$

The constant B_n includes the inversion constants at even weights and any terms with $k = 0$ at weight five. All remaining coefficients in (44)–(45) are exact rational row sums. The boundary computation is

n	u column			v column		
	E_n	B_n	$E_n - B_n$	E_n	B_n	$E_n - B_n$
2	7122	7522	-400	7695	7623	72
3	0	0	0	0	0	0
4	-2721/2	-1583	445/2	-7269/4	-7353/4	21
5	2205/8	597/8	201	-1011/4	-2357/8	335/8

For example, the weight-five rational-function identities themselves are

$$\sum_f u_f \mathcal{R}_5(f(t)) = \frac{2205}{8} \zeta(5), \quad \sum_f v_f \mathcal{R}_5(f(t)) = -\frac{1011}{4} \zeta(5), \quad 0 < t < 1. \quad (46)$$

The golden power with exponent zero contributes $597\zeta(5)/8$ in the first specialization and $-2357\zeta(5)/8$ in the second. Subtracting these terms gives $201\zeta(5)$ and $335\zeta(5)/8$, respectively. Multiplication by 144 and 120 therefore gives the weight-five constants 28944 and 5025.

For transparency, the power and sign bookkeeping in every weight can be expressed without any polylogarithm manipulation. A row of coefficient w has weighted coefficient wk^{5-n} . If $k < 0$ and n is even, add $2wk^{5-n}h_n(s)$ to B_n and negate the remaining coefficient. Replace k by $|k|$. If $s = -1$, replace its remaining coefficient z at $-\rho^k$ by $-z$ at ρ^k and $2^{1-n}z$ at ρ^{2k} . If $k = 0$, put the resulting constant into B_n . This finite procedure gives every entry above. Consequently

$$\sum_{j \geq 1} c_j j^{5-n} \mathcal{R}_n(\rho^j) = q_n \zeta(n), \quad 2 \leq n \leq 5, \quad (47)$$

with the constants stated in Theorem 3.1.

3.6 Removing the lower polylogarithms

The weight-one companion can be computed using the following elementary factorizations. Each equality follows by reduction modulo $\rho^2 + \rho - 1 = 0$, with $\sqrt{5} = 2\rho + 1$:

j	$1 - \rho^j$	j	$1 - \rho^j$
2	ρ	10	$11\rho^5$
4	$\sqrt{5}\rho^2$	12	$8\sqrt{5}\rho^6$
6	$4\rho^3$	20	$55\sqrt{5}\rho^{10}$
8	$3\sqrt{5}\rho^4$	24	$144\sqrt{5}\rho^{12}$

For either vector c , the coefficients of $\log 2$, $\log 3$, $\log 5$, and $\log 11$ in $\sum_j c_j j^4 \log(1 - \rho^j)$ vanish. Hence

$$S_1 = -\sum_j c_j j^4 \log(1 - \rho^j) = -\frac{1}{2} \sum_j c_j j^5 \lambda = a\lambda. \quad (48)$$

The two sums $\sum_j c_j j^5$ are -1440000 and 501120 , which give $a = 720000$ and $a = -250560$.

Proof of Theorem 3.1. The finite polynomial identities (42), Lemma 3.3, and the exact specialization arithmetic prove (47). Expand the definition of $\mathcal{R}_n$ at ρ^j . Since $\log(\rho^j) = j\lambda$, it gives

$$q_n \zeta(n) = \sum_{r=0}^{n-2} \frac{(-\lambda)^r}{r!} S_{n-r} + \frac{(-1)^{n-1}(n-1)}{n!} \lambda^{n-1} S_1. \quad (49)$$

Starting with (48), this triangular system has the unique solution (29). One may also verify the formula by substitution: the coefficients of each $q_j \zeta(j)$ reduce to

$$\sum_{r=0}^{n-j} \frac{(-1)^r}{r!(n-j-r)!} = \begin{cases} 1, & j = n, \\ 0, & j < n, \end{cases}$$

and the a term cancels because

$$\sum_{r=0}^{n-2} \frac{(-1)^r}{r!(n-r)!} + \frac{(-1)^{n-1}(n-1)}{n!} = 0.$$

This proves all five weights. Substituting $\zeta(2) = \pi^2/6$, $\zeta(4) = \pi^4/90$, and $\lambda = -\ell$ in the case $n = 5$ proves the two asserted evaluations. $\square$

Some useful lower-weight consequences, written with smaller integer coefficient vectors, are

$$5 \operatorname{Li}_2(\rho^2) - 15 \operatorname{Li}_2(\rho^4) - 2 \operatorname{Li}_2(\rho^{10}) + \operatorname{Li}_2(\rho^{20}) = 5\lambda^2 - \frac{2\pi^2}{15}, \quad (50)$$

$$50 \operatorname{Li}_3(\rho^2) - 75 \operatorname{Li}_3(\rho^4) - 4 \operatorname{Li}_3(\rho^{10}) + \operatorname{Li}_3(\rho^{20}) = \frac{100\lambda^3 - 8\pi^2\lambda}{3}, \quad (51)$$

$$500 \operatorname{Li}_4(\rho^2) - 375 \operatorname{Li}_4(\rho^4) - 8 \operatorname{Li}_4(\rho^{10}) + \operatorname{Li}_4(\rho^{20}) = \frac{500\lambda^4}{3} - \frac{80\pi^2\lambda^2}{3} + \frac{89\pi^4}{45}. \quad (52)$$

For the third displayed ladder one obtains

$$87 \operatorname{Li}_2(\rho^2) - 864 \operatorname{Li}_2(\rho^4) + 1152 \operatorname{Li}_2(\rho^6) + 864 \operatorname{Li}_2(\rho^8) - 144 \operatorname{Li}_2(\rho^{24}) = \pi^2 - 87\lambda^2, \quad (53)$$

$$\begin{aligned} 87 \operatorname{Li}_3(\rho^2) - 432 \operatorname{Li}_3(\rho^4) + 384 \operatorname{Li}_3(\rho^6) + 216 \operatorname{Li}_3(\rho^8) - 12 \operatorname{Li}_3(\rho^{24}) \\ = -58\lambda^3 + 2\pi^2\lambda, \end{aligned} \quad (54)$$

$$\begin{aligned} 87 \operatorname{Li}_4(\rho^2) - 216 \operatorname{Li}_4(\rho^4) + 128 \operatorname{Li}_4(\rho^6) + 54 \operatorname{Li}_4(\rho^8) - \operatorname{Li}_4(\rho^{24}) \\ = -29\lambda^4 + 2\pi^2\lambda^2 + \frac{7\pi^4}{90}. \end{aligned} \quad (55)$$

These companions are exact consequences of the same certificate, with no separate recognition of numerical constants.

3.7 The first ladder and canonical normalization

The first golden formula is included in the supplied replay data as a small consistency and normalization certificate. Its ordinary coefficient vector is

$$(c_2^I, c_4^I, c_6^I, c_{12}^I) = (4455, -1215, -360, 15).$$

A primitive functional certificate has the following twelve rows in the notation (36):

s	a	b	c	w	s	a	b	c	w
1	0	0	1	-54	1	1	0	0	-81
-1	0	0	1	-81	-1	1	0	0	-54
-1	0	0	3	1	-1	1	0	1	-9
1	1	0	-2	-9	-1	2	0	-1	-9
1	1	0	-1	-54	-1	3	0	-3	1
-1	1	0	-1	-81	1	3	0	0	1

Its scale factor is 240. The same calculation gives

$$(a, q_2, q_3, q_4, q_5) = (84240, -6480, 0, 3420, 3015),$$

recovering the previously proved first evaluation.

For completeness, define the manuscript's canonical ladders explicitly. For $N \in \{12, 20, 24\}$, write

$$T_N(m) = D_{N,1} \frac{\lambda^m}{m!} + D_{N,2} \zeta(2) \frac{\lambda^{m-2}}{(m-2)!} + D_{N,3} \zeta(4) \frac{\lambda^{m-4}}{(m-4)!},$$

with any term having a negative factorial index omitted, and put

$$\begin{aligned} L_{12}(m) &= \frac{\text{Li}_m(\rho^{12})}{12^{m-1}} - \frac{3 \text{Li}_m(\rho^6)}{2 \cdot 6^{m-1}} - \frac{\text{Li}_m(\rho^4)}{4^{m-1}} + \frac{11 \text{Li}_m(\rho^2)}{48 \cdot 2^{m-1}} + T_{12}(m), \\ L_{20}(m) &= \frac{\text{Li}_m(\rho^{20})}{20^{m-1}} - \frac{\text{Li}_m(\rho^{10})}{10^{m-1}} - 3 \frac{\text{Li}_m(\rho^4)}{4^{m-1}} + \frac{1 \text{Li}_m(\rho^2)}{2 \cdot 2^{m-1}} + T_{20}(m), \\ L_{24}(m) &= \frac{\text{Li}_m(\rho^{24})}{24^{m-1}} - \frac{4 \text{Li}_m(\rho^{12})}{3 \cdot 12^{m-1}} - 2 \frac{\text{Li}_m(\rho^8)}{8^{m-1}} \\ &\quad + \frac{7 \text{Li}_m(\rho^4)}{3 \cdot 4^{m-1}} - \frac{205 \text{Li}_m(\rho^2)}{576 \cdot 2^{m-1}} + T_{24}(m). \end{aligned}$$

The correction coefficients are

N	$D_{N,1}$	$D_{N,2}$	$D_{N,3}$
12	$-13/48$	$1/48$	$-19/1728$
20	$-1/2$	$1/25$	$-89/4000$
24	$179/576$	$-5/192$	$629/41472$

With these definitions, exact equality holds in

$$L_{12}(5) = \frac{67}{6912} \zeta(5), \quad L_{20}(5) = \frac{201}{10000} \zeta(5), \quad L_{24}(5) = -\frac{1541}{110592} \zeta(5).$$

For the last value, add $64/3$ times the first ordinary weight-five formula to the third, and divide by $-15 \cdot 24^4$; this eliminates $\text{Li}_5(\rho^6)$ and gives the manuscript's canonical index-24 combination. The corresponding normalized first-degree ladders vanish at weights 1, 2, 3, 4. These statements follow by exact rational recombination of (29) and its first-ladder counterpart.

3.8 Verification, integration, and further questions

The files `golden_rational_certificates.json` and `certify_golden_ladders.py` contain the complete finite data and an independent replay of the polynomial checks. The verifier uses SymPy to factor every rational function, retains rational-prime coordinates, checks the tensor equations directly in all four weights, performs every golden specialization, and checks the eight elementary factorizations used for S_1 . Its separate 85-digit calculations verify the rational-function identities at $t = 37/100$ and the ordinary companions at ρ ; these numerical diagnostics do not enter the proof. The detailed receipt is `golden_certificate_verification.json`.

For manuscript integration, the second and third displayed weight-five formulas can be marked proved and linked to this section. Their first-degree lower-weight vanishing and the canonical values just displayed can be marked proved as well. The weight-6 through weight-9 reconstructions have a different status: the present certificates establish no upward continuation beyond weight five. The contraction in Lemma 3.3 decreases weight; there is no inverse contraction that would produce the missing higher functional identity.

Several follow-up questions are concrete. First, can the 58- and 60-term certificates be decomposed into a short list of named Kummer specializations, giving a smaller human derivation? Second, can a larger rational-function alphabet produce the manuscript's weight-6 and weight-7 golden identities

with similarly complete endpoint bookkeeping? Third, which part of the golden relation space is obtained from a prescribed degree bound, and can exact matrix-rank certificates prove useful minimality statements within that finite search model? Fourth, the rational-function identities (46) can be specialized at other algebraic values satisfying multiplicative relations among $t, 1 - t, 1 + t$; determining when their descended constants close on a small polylogarithm basket is a systematic route to additional ladders.

4 A certified double radial turning-point bifurcation

The local radial problem has a second layer of degeneracy. The manuscript proves that the first nonconstant coefficient of the normalized radius changes sign at a unique outer-order threshold for each inner order $0 < b < 1$. It subsequently proves opposite signs of the next coefficient at two points of that threshold curve, and locates a possible transition numerically [1]. Its final question in `05-real-turning.tex` asks for the higher-order behavior at such a transition. We certify one transition, determine its next coefficient, and prove that nearby normalized-radius curves can have two small-radius turning points: a minimum followed by a maximum.

The conclusion is local in both the parameters and the radius. It does not settle the number of quartic transitions over the entire interval $0 < b < 1$, nor exclude additional turning points farther from the origin. These distinctions matter because an initial Taylor coefficient, even with a proved sign, does not control the whole disk.

4.1 An exact finite recurrence to every radial order

For $a, b > 0$, put

$$F_{a,b}(z) = \sum_{n=2}^{\infty} p_n(a, b) z^n, \quad p_n(a, b) = \frac{H_{n-1}^{(b)}}{n^a}, \quad H_{n-1}^{(b)} = \sum_{j=1}^{n-1} j^{-b}. \quad (56)$$

Let $\theta_{a,b}(\rho)$ denote the local solution near $\pi/2$ of

$$\operatorname{Im} F_{a,b}(\rho e^{i\theta}) = 0,$$

for positive sufficiently small ρ , and set

$$\eta_{a,b}(\rho) = \frac{\cos \theta_{a,b}(\rho)}{\rho}, \quad t = \rho^2. \quad (57)$$

The radius ρ in this section is a disk radius. We write the analytic extension in the squared radius as $\eta_{a,b}(\rho) = \Phi(a, b, t)$.

Proposition 4.1 (An all-orders finite coefficient identity). *For $m \geq 0$, define the polynomial*

$$E_m(y) = \sum_{n=m+2}^{2m+3} (-1)^{n-m-2} \binom{m+1}{n-m-2} 2^{2m+3-n} p_n y^{2m+3-n}. \quad (58)$$

Locally in (a, b, t) , the normalized-radius equation is the convergent analytic identity

$$\sum_{m=0}^{\infty} t^m E_m(\Phi(a, b, t)) = 0. \quad (59)$$

There is a unique local analytic solution with $\Phi(a, b, 0) = \mu := p_3/(2p_2)$. If

$$\Phi(a, b, t) = \mu + \sum_{j \geq 1} c_j(a, b) t^j,$$

then the exact recursive identity is

$$c_j = -\frac{1}{2p_2} [t^j] \sum_{m=1}^j t^m E_m \left(\mu + \sum_{\ell=1}^{j-1} c_\ell t^\ell \right). \quad (60)$$

In particular, c_j depends only on $p_2, \dots, p_{2j+3}$.

Proof. For θ near $\pi/2$, divide the imaginary-part equation by $\rho^3 \sin \theta$ and use $\sin(n\theta) = \sin \theta U_{n-1}(\cos \theta)$. The finite Chebyshev identity is

$$U_{n-1}(x) = \sum_{h=0}^{\lfloor (n-1)/2 \rfloor} (-1)^h \binom{n-1-h}{h} (2x)^{n-1-2h}.$$

After substituting $x = \rho y$, a summand in $p_n \rho^{n-3} U_{n-1}(\rho y)$ has radius exponent $2n - 4 - 2h$. It contributes to t^m exactly when $h = n - 2 - m$. The inequalities $0 \leq h \leq \lfloor (n-1)/2 \rfloor$ are equivalent to $m + 2 \leq n \leq 2m + 3$. Substitution gives precisely (58).

This is an analytic identity, not merely a formal rearrangement. On a compact complex neighborhood of fixed positive (a, b) , the coefficients $p_n(a, b)$ are analytic and grow at most polynomially in n . The Chebyshev recurrence bounds $U_{n-1}(\rho y)$ exponentially in n , uniformly when y stays bounded and ρ is sufficiently small. Hence the normalized series converges locally uniformly. The apparent negative power in its $n = 2$ term cancels, leaving $2p_2 y$; all remaining powers of ρ are nonnegative and even. The sum is therefore jointly analytic in (a, b, t, y) .

Now $E_0(y) = 2p_2 y - p_3$, and $2p_2 \neq 0$ at real positive parameters. The analytic implicit-function theorem gives the unique solution $y = \Phi(a, b, t)$ through $\mu = p_3/(2p_2)$. For small positive ρ , the angle $\arccos(\rho \Phi(a, b, \rho^2))$ is near $\pi/2$ and solves the original equation, proving the claimed local interpretation. Finally, the contribution of E_0 to the coefficient of t^j is $2p_2 c_j$. Since every other term has a factor t^m with $m \geq 1$, its coefficient of degree j uses only the already determined $c_1, \dots, c_{j-1}$. Solving for c_j proves (60). $\square$

The recurrence makes the finite nature of a local degeneracy test explicit. For example, no coefficient of $F_{a,b}$ beyond p_9 is needed to determine the first three nonconstant coefficients of η . This is useful when a small coefficient results from nearly cancelling terms: one can certify a finite expression directly instead of estimating derivatives from values of the full boundary function.

4.2 The sextic coefficient

Write

$$\Phi(a, b, t) = \mu + Kt + Qt^2 + Rt^3 + O(t^4). \quad (61)$$

Thus K, Q, R are respectively the coefficients of ρ^2, ρ^4, ρ^6 in the normalized radius. Define

$$\begin{aligned} A(y) &= 4p_3 y^2 - 4p_4 y + p_5, \\ B(y) &= 8p_4 y^3 - 12p_5 y^2 + 6p_6 y - p_7, \\ C(y) &= 16p_5 y^4 - 32p_6 y^3 + 24p_7 y^2 - 8p_8 y + p_9. \end{aligned} \quad (62)$$

These are E_1, E_2, E_3 in Proposition 4.1.

Proposition 4.2 (The first three nonconstant coefficients). *The exact coefficient formulas are*

$$\begin{aligned} K &= -\frac{A(\mu)}{2p_2}, \\ Q &= -\frac{A'(\mu)K + B(\mu)}{2p_2}, \\ R &= -\frac{A'(\mu)Q + 4p_3K^2 + B'(\mu)K + C(\mu)}{2p_2}. \end{aligned} \quad (63)$$

At any simultaneous zero of K and Q ,

$$R = -\frac{C(\mu)}{2p_2}. \quad (64)$$

All these quantities are real analytic in positive (a, b) .

Proof. Retaining degrees at most three in (59) gives

$$0 = 2p_2\Phi - p_3 + tA(\Phi) + t^2B(\Phi) + t^3C(\Phi) + O(t^4).$$

The coefficients of t, t^2, t^3 are respectively

$$\begin{aligned} &2p_2K + A(\mu), \\ &2p_2Q + A'(\mu)K + B(\mu), \\ &2p_2R + A'(\mu)Q + \tfrac{1}{2}A''(\mu)K^2 + B'(\mu)K + C(\mu). \end{aligned}$$

Since $A'' = 8p_3$, their vanishing gives (63). The denominators do not vanish at positive parameters, so analyticity follows from the finite formulas. $\square$

The manuscript already identifies K and Q and proves that, on the curve $K = 0$, the quartic coefficient has opposite signs at $b = 1/2$ and $b = 999/1000$. The sextic formula supplies the missing test when both of those earlier coefficients vanish.

4.3 A unique certified transition in a rational rectangle

Let $\mathcal{B}$ be the open rectangle with rational endpoints

$$\begin{aligned} a_- &= 1.0014301809614951381293517293, \\ a_+ &= 1.0014301809614951381293517295, \\ b_- &= 0.9974937898734204210746055930669, \\ b_+ &= 0.9974937898734204210746055930670. \end{aligned} \quad (65)$$

The many digits here specify an exact, reproducible rational box. They do not play the role of unproved floating-point approximations.

Theorem 4.3 (A nondegenerate quartic transition). *There is exactly one pair $(a_{\dagger}, b_{\dagger}) \in \mathcal{B}$ such that $K(a_{\dagger}, b_{\dagger}) = Q(a_{\dagger}, b_{\dagger}) = 0$. Let $R_0 = R(a_{\dagger}, b_{\dagger})$. Then*

$$\begin{aligned} -3.368744535599044 \cdot 10^{-11} &< R_0 < -3.368744535599042 \cdot 10^{-11}, \\ -8.315139843259675 \cdot 10^{-10} &< \det D(K, Q) < -8.315139843259673 \cdot 10^{-10}. \end{aligned} \quad (66)$$

The determinant in the second line is evaluated at the certified pair. In particular, $(a, b) \mapsto (K, Q)$ is a local real analytic diffeomorphism there.

The curve $K = 0$ is locally a graph $a = a_*(b)$ throughout the b interval in (65). If $q_*(b) = Q(a_*(b), b)$, then its zero is unique in that interval, and

$$4.8702847165970496 \cdot 10^{-8} < q'_*(b_+) < 4.8702847165970497 \cdot 10^{-8}. \quad (67)$$

Proof. The accompanying program `code/certify_double_turning.py` uses only integers and rational numbers. Its exact interval output is `data/double_turning_certificate.json`. We first describe the elementary arithmetic that makes the certificate a finite proof, and then explain how its inequalities imply the theorem.

Every interval operation is rounded outward to multiples of 2^{-256} . For integers $1 \leq n \leq 9$, write $z = (n - 1)/(n + 1)$ and use

$$\log n = 2 \sum_{j \geq 0} \frac{z^{2j+1}}{2j+1}.$$

After the first $L = 440$ terms, the omitted positive tail is at most

$$\frac{2z^{2L+1}}{(2L+1)(1-z^2)}. \quad (68)$$

For a rational $0 \leq x < 3$, the Taylor polynomial for e^x through degree $M = 90$ has omitted positive tail at most

$$\frac{x^{M+1}}{(M+1)!} \frac{1}{1 - x/(M+2)}. \quad (69)$$

These bounds follow respectively by bounding the remaining denominators from below and the ratio of consecutive omitted terms from above. All exponential arguments occurring in n^{-a} and j^{-b} on the specified boxes are less than three. Monotonicity, the displayed bounds, and reciprocation therefore enclose every needed coefficient $p_2, \dots, p_9$.

The first derivatives in a and b are propagated by the exact sum, product, reciprocal, integer-power, and exponential chain rules. Thus the derivative enclosures do not use finite differences. Substituting the intervals into (63) yields, throughout the closed rectangle,

$$K_a < 0, \quad K_b < 0, \quad K_a Q_b - K_b Q_a < 0,$$

and the intervals in (66) and (67). In the last case the expression enclosed is $(K_a Q_b - K_b Q_a)/K_a$. For the value of R at a simultaneous zero, the program additionally evaluates the simpler expression (64).

The two vertical face estimates, again valid over the full b interval, include the strict rational inequalities

$$K(a_-, b) > 1.45 \cdot 10^{-30}, \quad K(a_+, b) < -1.84 \cdot 10^{-30}. \quad (70)$$

The intermediate value theorem and $K_a < 0$ give exactly one root $a = a_*(b)$ between the vertical faces for every such b . The implicit-function theorem makes this graph real analytic.

At $b = b_-$ and $b = b_+$ separately, the certificate encloses this root in rational a intervals of width 10^{-60} . It verifies positive and negative signs of K at their left and right endpoints and encloses Q on each whole interval. The resulting strict bounds are

$$\begin{aligned} -2.654 \cdot 10^{-39} &< q_*(b_-) < -2.653 \cdot 10^{-39}, \\ 2.216 \cdot 10^{-39} &< q_*(b_+) < 2.217 \cdot 10^{-39}. \end{aligned} \quad (71)$$

The tiny magnitudes are the reason to use outward exact arithmetic: their signs would not be established by an unqualified decimal evaluation. The complete rational endpoints are preserved in the machine-readable certificate.

Differentiating $K(a_*(b), b) = 0$ gives

$$q'_*(b) = Q_b - \frac{K_b}{K_a} Q_a = \frac{K_a Q_b - K_b Q_a}{K_a} > 0 \quad (72)$$

throughout the rectangle. The signs in (71) therefore give exactly one simultaneous zero. At that zero, (64) and its negative interval prove the first line of (66); the determinant interval proves the local invertibility of the parameter map. $\square$

Remark 4.4 (The exact scope of uniqueness). The theorem certifies uniqueness in $\mathcal{B}$, equivalently along the specified short interval of the threshold graph. The predecessor proves existence of a quartic transition in a much larger interval. Neither statement, nor their combination, proves that no additional quartic transition occurs elsewhere in $0 < b < 1$. The small rectangle locates a genuine transition and determines its local type.

4.4 The local fold and the region with two extrema

The nonzero determinant has a concrete consequence: the quadratic and quartic coefficients themselves are valid local parameter coordinates. Write

$$k = K(a, b), \quad q = Q(a, b),$$

and use the analytic inverse map to regard Φ as a function of (k, q, t) . The symbol q here is this local coefficient coordinate, whereas $q_*(b)$ in (72) denotes its restriction to the threshold graph.

Theorem 4.5 (Complete local turning-point classification). *There exist $T > 0$, a neighborhood of $(k, q) = (0, 0)$, and a real analytic function $\kappa(q)$ such that*

$$\kappa(0) = \kappa'(0) = 0, \quad \kappa(q) = \frac{q^2}{3R_0} + O(q^3). \quad (73)$$

For positive sufficiently small q , the following statements count all critical points of $\eta(\rho)$ in the fixed interval $0 < \rho < \sqrt{T}$:

1. *If $\kappa(q) < k < 0$, there are exactly two critical points, a strict local minimum followed by a strict local maximum.*
2. *If $k = \kappa(q)$, there is exactly one degenerate stationary point. It satisfies $\eta' = \eta'' = 0$ and $\eta''' < 0$ and is not a local extremum.*
3. *If $k < \kappa(q)$, there are no critical points.*
4. *If $k \geq 0$, there is exactly one critical point, a strict local maximum.*

For $q \leq 0$, there is exactly one strict local maximum if $k > 0$, and there are no critical points if $k \leq 0$. All assertions concern parameters in the stated neighborhood.

Proof. Set

$$U(k, q, t) = \partial_t \Phi(k, q, t) = k + 2qt + 3R(k, q)t^2 + O(t^3). \quad (74)$$

The remainder is jointly analytic, and $U_{tt}(0, 0, 0) = 6R_0 < 0$. Choose $T > 0$ sufficiently small and then shrink the parameter neighborhood so that

$$U_{tt}(k, q, t) < 0 \quad (0 \leq t \leq T), \quad U(k, q, T) < 0. \quad (75)$$

This is possible because $U(0, 0, t) = 3R_0 t^2 + O(t^3) < 0$ for all sufficiently small positive t . In particular, U is strictly concave as a function of t throughout this fixed interval.

The analytic implicit-function theorem applied to $U_t = 0$ gives a unique local analytic solution $t = t_c(k, q)$ near the origin. Since $U_t(k, 0, 0) = 0$ identically, uniqueness gives $t_c(k, 0) = 0$. Its expansion is

$$t_c(k, q) = -\frac{q}{3R(k, 0)} + O(q^2). \quad (76)$$

After another restriction of the parameter neighborhood, $R(k, 0) < 0$, and hence $0 < t_c(k, q) < T$ when $q > 0$. By strict concavity this point is the unique maximum of U on $[0, T]$.

Let $H(k, q) = U(k, q, t_c(k, q))$. At the origin, $H = 0$ and $H_k = 1$. The implicit-function theorem gives a unique analytic zero curve $k = \kappa(q)$, and we may assume $H_k > 0$ throughout the neighborhood. Also $H_q(0, 0) = 0$, so $\kappa'(0) = 0$ and $\kappa(q) = O(q^2)$. Using (76) on this curve yields

$$t_c(\kappa(q), q) = -\frac{q}{3R_0} + O(q^2),$$

and substitution into (74) gives

$$0 = H(\kappa(q), q) = \kappa(q) - \frac{q^2}{3R_0} + O(q^3).$$

This proves (73); in particular, $\kappa(q) < 0$ for small positive q .

Suppose first that $q > 0$ and $\kappa(q) < k < 0$. The function U is negative at zero, positive at its unique maximum, and negative at T . Its strict concavity makes it strictly increasing before its maximum and strictly decreasing after it. It has exactly two zeros, the first with $U_t > 0$ and the second with $U_t < 0$. Since

$$\eta'(\rho) = 2\rho U(k, q, \rho^2), \quad (77)$$

they give a strict minimum and a strict maximum, respectively. At a simple zero, this can also be read from $\eta''(\rho) = 4\rho^2 U_t$.

If $k = \kappa(q)$, the maximum of U is zero, so it is the only zero. At its corresponding positive radius, $U = U_t = 0$ and $U_{tt} < 0$. Differentiation of (77) gives

$$\eta'''(\rho) = 12\rho U_t + 8\rho^3 U_{tt} = 8\rho^3 U_{tt} < 0,$$

while $\eta' = \eta'' = 0$. The derivative of η is negative on both sides, so this stationary point is not an extremum. If $k < \kappa(q)$, the maximum of U is negative and there are no zeros. If $k \geq 0$, U starts nonnegative, initially increases when $q > 0$, and then decreases to its negative value at T . It has exactly one positive zero, a strict maximum of η .

Finally, when $q \leq 0$, one has $U_t(k, q, 0) = 2q \leq 0$ and $U_{tt} < 0$. Thus U is strictly decreasing for $0 < t \leq T$. Its initial value is k , and its value at T is negative; this proves the remaining alternatives. $\square$

The pair of critical points is born on the curve $k = \kappa(q)$ and one reaches the endpoint $t = 0$ on the curve $k = 0$. These two parameter curves are tangent at the certified transition. The proof is a quadratic fold of stationary points in the variable t , with the additional physical boundary $t = \rho^2 \geq 0$. No terminology from catastrophe theory is needed to infer the count or the types of the critical points: they follow from the strict concavity in (75).

4.5 The sextic curve and two radius scaling laws

At the transition itself, the earlier local tests are both zero, but the normalized radius is not locally constant.

Corollary 4.6 (Behavior at the transition). *For the certified pair,*

$$\eta_{a_{\dagger}, b_{\dagger}}(\rho) = \mu_{\dagger} + R_0 \rho^6 + O(\rho^8), \quad R_0 < 0. \quad (78)$$

It decreases strictly for all sufficiently small positive radii.

Proof. The expansion follows from $K = Q = 0$ and Proposition 4.2. Analyticity permits differentiation, giving $\eta'(\rho) = 6R_0\rho^5 + O(\rho^7) < 0$ for small positive ρ . $\square$

Corollary 4.7 (Separation of the two turning radii). *Fix $0 < \lambda < 1$. For sufficiently small $\varepsilon > 0$, use the local inverse parameter map to impose*

$$q = -3R_0\varepsilon, \quad k = 3R_0\lambda\varepsilon^2. \quad (79)$$

The resulting parameters have two local turning points, with radii

$$\begin{aligned} \rho_{\min}(\varepsilon) &= \sqrt{\varepsilon} \sqrt{1 - \sqrt{1 - \lambda}} + O(\varepsilon^{3/2}), \\ \rho_{\max}(\varepsilon) &= \sqrt{\varepsilon} \sqrt{1 + \sqrt{1 - \lambda}} + O(\varepsilon^{3/2}). \end{aligned} \quad (80)$$

The error estimates are uniform when λ ranges over a compact subinterval of $(0, 1)$.

Proof. Here $q > 0$ and $k < 0$. Equation (73) gives

$$\kappa(q) = 3R_0\varepsilon^2 + O(\varepsilon^3),$$

so $\kappa(q) < k < 0$ for fixed $0 < \lambda < 1$ and sufficiently small ε . Theorem 4.5 applies. To determine the radii, set $t = \varepsilon x$ in (74). The inverse parameter map is analytic, hence $R(k, q) = R_0 + O(\varepsilon)$ along (79). Dividing by $3R_0\varepsilon^2$ gives

$$0 = x^2 - 2x + \lambda + O(\varepsilon).$$

The two limiting roots $1 \pm \sqrt{1 - \lambda}$ are positive and simple. The analytic implicit-function theorem gives $x_{\pm} = 1 \pm \sqrt{1 - \lambda} + O(\varepsilon)$, and taking positive square roots proves (80). On a compact subinterval of $(0, 1)$, the roots stay separated from one another and from zero, giving the stated uniformity. $\square$

Corollary 4.8 (A fourth-root law on the quartic-zero curve). *Along the local curve $q = 0$, for sufficiently small $k > 0$, the unique local maximum satisfies*

$$\rho_{\max}(k, 0) = \left(\frac{k}{-3R_0} \right)^{1/4} + O(k^{3/4}). \quad (81)$$

Proof. Theorem 4.5 gives the unique local maximum. Since $R(k, 0) = R_0 + O(k)$, substitute $t = \sqrt{k}x$ in $U = 0$ and divide by k . The equation becomes

$$0 = 1 + 3R_0x^2 + O(\sqrt{k}).$$

Its positive limiting root is $(-3R_0)^{-1/2}$ and is simple. The implicit-function theorem, with $\sqrt{k}$ as the small parameter, gives $t = \sqrt{k}(-3R_0)^{-1/2} + O(k)$. Taking the positive square root yields (81). $\square$

The square-root splitting in (80) and the fourth-root law in (81) concern specified analytic paths in the original (a, b) parameter space. They are not fits to numerical extrema. The existence of those paths follows from the certified nonzero determinant, and their exponents follow from the proved local equation.

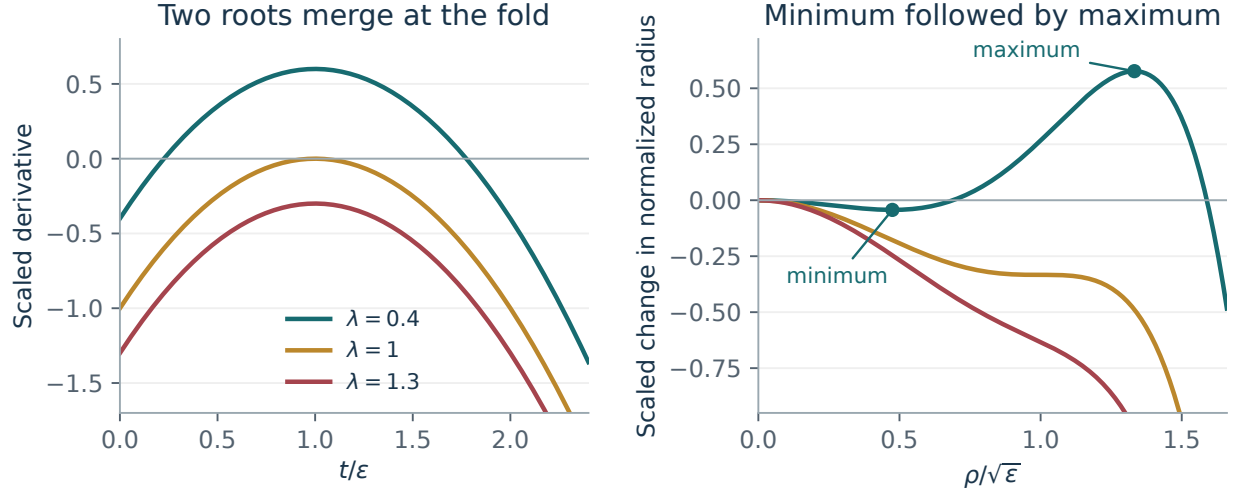


Figure 2: The limiting local model on the paths (79). With $z = t/\varepsilon$ and $u = \rho/\sqrt{\varepsilon}$, the scaled derivative is $-\lambda + 2z - z^2$ and the scaled change of normalized radius is $-\lambda u^2 + u^4 - u^6/3$. The latter uses the positive scale $-3R_0\varepsilon^3$. These are the proved leading local terms, not direct numerical evaluations of the full function. The exact fold has higher-order corrections to the displayed $\lambda = 1$ model.

4.6 What the result resolves and what remains

The manuscript's displayed quartic-transition location is now replaced by an exact rational enclosure, a local uniqueness theorem, a transversality sign, and a negative sextic coefficient. Its request for the local higher-order radial behavior at that transition is answered by Theorem 4.5. The resulting region with a minimum followed by a maximum is stronger than the preceding existence results for one minimum or one maximum separately.

Three natural questions remain distinct. First, is this the only zero of $Q(a_*(b), b)$ in $0 < b < 1$? Second, can one give explicit numerical lower bounds for the parameter neighborhood and the fixed radius $\sqrt{T}$ in the fold theorem? Third, do the same parameter pairs have additional turning points at larger radii? The present finite certificate answers none of these global or quantitative questions by itself. It establishes a nondegenerate local mechanism that any proposed global classification must accommodate.

Another useful direction is the fully constant point $(a, b) = (1, 1)$, where $F_{1,1}(z) = \frac{1}{2} \log^2(1 - z)$ and $\eta_{1,1}(\rho) = 1/2$. At that point every nonconstant radial coefficient vanishes. Relating that higher degeneracy to the nearby certified fold requires further analysis; it cannot be inferred from the sign of R_0 alone. Proposition 4.1 provides an exact finite calculation at each order for such an investigation.

5 A constructive normal form for the full Cayley shuffle ideal

The earlier Cayley report determines the dimensions of two formal presentations: the span of the unmultiplied Cayley rows, and the larger shuffle ideal generated by them. Its weight-seven search notes identify an explicit reconstruction map for the latter quotient as a missing computational step. We provide such a map in every weight. It gives exact representatives without a relation matrix and produces a concrete obstruction for the frozen S_6 formula after *all* shuffle multiples of Cayley

identities have been included.

The shuffle-polynomial theorem and Eulerian idempotents used below are classical [7, 8]. The contribution to the inspected project is the compatible endpoint projection, its combination with those tools, the explicit reconstruction formula, and the resulting weight-seven certificate. Neither a period-independence assertion nor the conjectured S_6 equality is established here.

5.1 The algebra and its precise relation ideal

Use the outer-letter-first convention and the five letters

$$z = \frac{dt}{t}, \quad c = \frac{dt}{1-t}, \quad a = \frac{i dt}{1-it}, \quad b = -\frac{dt}{1+t}, \quad \bar{a} = -\frac{i dt}{1+it}.$$

Let $\tilde{\mathcal{A}}$ be the rational shuffle algebra on these letters, with empty word 1 and shuffle product $\sqcup\sqcup$. Write $\mathcal{A}$ for its subalgebra spanned by 1 and all words with first letter different from c and last letter different from z . For such a word w , $H(w)$ is its convergent iterated integral from zero to one. Products written inside a word are concatenations unless the shuffle symbol is displayed.

The Cayley involution is

$$C(v_1 \cdots v_n) = \tau(v_n) \cdots \tau(v_1), \quad \frac{v}{\tau(v)} \mid \frac{z}{c-b} \frac{c}{z+b} \frac{a}{b-\bar{a}} \frac{b}{b} \frac{\bar{a}}{b-a}.$$

It preserves $\mathcal{A}$, is an automorphism for the shuffle product, and commutes with the conjugation K that exchanges $a, \bar{a}$ and fixes z, c, b . The change of variable $t \mapsto (1-t)/(1+t)$ proves $H(Cw) = H(w)$ for admissible w . Define

$$\mathcal{J} = \langle f - Cf : f \in \mathcal{A} \rangle_{\text{shuffle}}. \quad (82)$$

Thus $\mathcal{J}$ includes arbitrary products of convergent Cayley relations with admissible words. No shuffle, distribution, or other functional equation is imposed in this definition.

5.2 Endpoint projection that commutes with Cayley symmetry

The ordinary projection that sets the divergent shuffle generators z, c to zero is unsuitable: it does not commute with C . For example, the ordinary projection of $Cz = c - b$ is $-b$, whereas C applied to the projection of z is zero. The needed correction is small but essential. Put

$$x = z + \frac{b}{2}, \quad y = c - \frac{b}{2}. \quad (83)$$

Then $Cx = y$, $Cy = x$, and $Kx = x$, $Ky = y$.

Lemma 5.1 (Equivariant endpoint projection). *There is a unique shuffle-algebra retraction $\mathcal{R} : \tilde{\mathcal{A}} \rightarrow \mathcal{A}$ that is the identity on $\mathcal{A}$ and sends x, y to zero. It commutes with C, K . Equivalently, it substitutes $z = -b/2$, $c = b/2$ in shuffle-polynomial coordinates.*

For a word w of length n , let l be its initial run of c 's and r its terminal run of z 's. If $w[u : n-v]$ denotes deletion of u initial letters and v terminal letters, then

$$\mathcal{R}(w) = \sum_{u=0}^l \sum_{v=0}^r \left(\frac{b}{2} - c\right)^u \sqcup\sqcup w[u : n-v] \sqcup\sqcup \left(-\frac{b}{2} - z\right)^v. \quad (84)$$

The powers in this formula are concatenation powers of linear forms.

Proof. Order the alphabet as $z < a < b < \bar{a} < c$. Every Lyndon word of length at least two is admissible. The shuffle-polynomial theorem [7] and the triangular Lyndon factorization therefore give $\tilde{\mathcal{A}} = \mathcal{A}[z, c] = \mathcal{A}[x, y]$ as polynomial algebras. Setting x, y to zero defines the claimed retraction. The ideal (x, y) is stable under both C, K , and both automorphisms preserve $\mathcal{A}$; hence the retraction commutes with them.

For the formula, deletion of one initial c and deletion of one terminal z are commuting derivations of the shuffle algebra. Their Taylor substitutions replace c by $b/2$ and z by $-b/2$. The u th shuffle power of a single linear form is $u!$ times its concatenation power, which cancels the Taylor factorial. This yields (84). In particular every term left after cancellation is admissible; this also follows from the range of the retraction. $\square$

This is a formal polynomial projection. No value is assigned to a divergent iterated integral, and no regularized analytic identity is being assumed. For example,

$$\mathcal{R}(az) = -za - \frac{ba + ab}{2}, \quad \mathcal{R}(c) = \frac{b}{2}, \quad \mathcal{R}(z) = -\frac{b}{2}.$$

The first formula also illustrates why a literal letter substitution inside concatenation words would be incorrect.

5.3 Explicit eigenvector lifts and the normal-form theorem

For a nonempty word w , define the first Eulerian operator by

$$e(w) = \sum_{k=1}^{|w|} \frac{(-1)^{k-1}}{k} \sum_{\substack{w=w_1 \cdots w_k \\ |w_j| > 0}} w_1 \sqcup \cdots \sqcup w_k, \quad e(1) = 0. \quad (85)$$

The inner sum is over consecutive nonempty blocks. This is the convolution logarithm of the identity for shuffle multiplication and deconcatenation. It annihilates a shuffle product of two positive-weight elements and satisfies

$$e^2 = e, \quad e(f) - f \in (\tilde{\mathcal{A}}_{>0})^2 \quad (f \in \tilde{\mathcal{A}}_{>0}). \quad (86)$$

These are the standard Eulerian-idempotent properties in a connected commutative Hopf algebra [8]. The congruence is immediate from (85); together with annihilation of products it also proves the displayed idempotence. Reversing the consecutive blocks and using commutativity of shuffle shows directly that e commutes with C ; it also commutes with K .

Set

$$E = \mathcal{R}e, \quad E_+(w) = \frac{E(w) + CE(w)}{2}. \quad (87)$$

All values of E, E_+ are admissible word polynomials. The following formula contains only finite sums of explicitly given rational words.

Theorem 5.2 (Constructive full-ideal projector). *Define $\Pi(1) = 1$ and, for every nonempty word,*

$$\Pi(w) = \sum_{k=1}^{|w|} \frac{1}{k!} \sum_{\substack{w=w_1 \cdots w_k \\ |w_j| > 0}} E_+(w_1) \sqcup \cdots \sqcup E_+(w_k). \quad (88)$$

Then $\Pi : \tilde{\mathcal{A}} \rightarrow \mathcal{A}$ is a graded shuffle-algebra homomorphism. Its restriction to $\mathcal{A}$ satisfies

$$\Pi^2 = \Pi, \quad \Pi C = C\Pi = \Pi, \quad \Pi K = K\Pi, \quad \ker(\Pi|_{\mathcal{A}}) = \mathcal{J}. \quad (89)$$

Consequently $\Pi(f) = \Pi(g)$ if and only if $f - g \in \mathcal{J}$ for $f, g \in \mathcal{A}$. In particular, every admissible word has the proved period identity

$$H(w) = H(\Pi(w)). \quad (90)$$

Proof. First restrict E to $\mathcal{A}$. It annihilates products of two positive-weight elements, and $E(f) - f \in (\mathcal{A}_{>0})^2$ because $\mathcal{R}$ preserves products and positive degree. Hence $E^2 = E$ on $\mathcal{A}$, and its image $U = E(\mathcal{A})$ maps isomorphically to the indecomposables $\mathcal{A}_{>0}/(\mathcal{A}_{>0})^2$. Lifts of any homogeneous basis of those indecomposables freely generate the polynomial algebra $\mathcal{A}$: the change from its Lyndon generators is degree triangular with invertible linear part. Thus multiplication gives an isomorphism $\text{Sym}(U) \simeq \mathcal{A}$.

Since $\mathcal{R}, e$ commute with C, K , the space U is stable under both. Write $U = U_+ \oplus U_-$ for its C eigenspaces. The algebra map that fixes U_+ and kills U_- is an idempotent map P on $\mathcal{A}$. Its kernel is the ideal generated by U_- . This ideal equals $\mathcal{J}$: if $u \in U_-$, then $u - Cu = 2u$; conversely C acts as the identity after the negative generators have been killed, so every $f - Cf$ belongs to that ideal. The map P commutes with K .

It remains to verify that the fully explicit formula is this map. For every nonempty w , convolution exponentiation gives

$$w = \sum_{k=1}^{|w|} \frac{1}{k!} \sum_{w=w_1 \cdots w_k} e(w_1) \sqcup \cdots \sqcup e(w_k).$$

Applying $\mathcal{R}$ gives the same formula for $\mathcal{R}(w)$ with factors $E(w_j)$. Those factors belong to U , including when the blocks themselves are not admissible. To see this last point, write $\tilde{\mathcal{A}} = \mathcal{A}[x, y]$. In degree at least two, every term of $w - \mathcal{R}(w)$ that involves x or y is a product of positive-degree factors and is annihilated by e ; in degree one, applying $\mathcal{R}e$ kills x, y directly. Thus $E(w) = E(\mathcal{R}(w)) \in U$ in every degree.

Now apply the algebra map P . On U , it acts by $(1+C)/2$, so the result is exactly (88). Therefore $\Pi = P\mathcal{R}$, proving multiplicativity and every assertion in (89). Finally, $w - \Pi(w) \in \mathcal{J}$ for admissible w . The map H annihilates this ideal by the convergent Cayley identity and the shuffle product law, proving (90). $\square$

For example, the three admissible weight-one letters give

$$\Pi(a) = \frac{a + b - \bar{a}}{2}, \quad \Pi(\bar{a}) = \frac{\bar{a} + b - a}{2}, \quad \Pi(b) = b.$$

Evaluating the first identity recovers $\text{Li}_1(i) + \text{Li}_1(-i) = \text{Li}_1(-1)$, including its sign. At higher weights the projector automatically includes the products of the corresponding Cayley-negative generators, which the raw average $(1+C)/2$ does not annihilate. The image is the same quotient whose weight-seven imaginary dimension was previously proved to be 7518. Formula (88) supplies representatives in the original word coordinates; it does not claim that the resulting sparse supports are already a chosen 7518-element basis.

5.4 An exact obstruction for the frozen weight-seven target

The series convention is $\text{Li}_{r,s}(u, v) = \sum_{n>m \geq 1} u^n v^m / (n^r m^s)$. Consequently

$$H(z^{r-1} a z^{s-1} a) = \text{Li}_{r,s}(i, 1), \quad H(z^5 a \bar{a}) = \text{Li}_{6,1}(i, -1).$$

Here $G = \beta(2)$ is Catalan's constant, and

$$S_6 = \sum_{n=0}^{\infty} \frac{(-1)^n H_n}{(2n+1)^6}, \quad H_n = \sum_{j=1}^n \frac{1}{j}, \quad H_0 = 0, \quad g_{r,s} = \text{Im Li}_{r,s}(i, 1).$$

Let F be the following explicitly fixed formal lift of the conjectured integer relation:

$$\begin{aligned} F = & -179712000z^5aa + 485683200z^5a\bar{a} - 36864000z^3az^2a + 401080320zaz^4a \\ & - \frac{47600087040}{61}z^6a + 11750400(za \sqcup z^4c) \\ & + 109347840(z^3a \sqcup z^2c) - 971366400(z^5a \sqcup b). \end{aligned} \quad (91)$$

Indeed $\beta(7) = 61\pi^7/184320$, $\text{Li}_1(-1) = -\log 2$, and the harmonic parity split gives

$$\begin{aligned} \text{Im } H(F) = & 485683200S_6 - 665395200g_{6,1} - 36864000g_{4,3} + 401080320g_{2,5} \\ & - 258247\pi^7 + 11750400G\zeta(5) + 109347840\beta(4)\zeta(3) + 971366400\beta(6)\log 2. \end{aligned}$$

For a word polynomial f , define its imaginary coordinate reduction $\mathcal{O}(f)$ by replacing each conjugate pair w, Kw by $(f_w - f_{Kw})w$, choosing the lexicographically smaller representative under the code order $z < c < a < b < \bar{a}$; conjugation-fixed words are discarded. This convention encodes imaginary parts without a factor of $1/2$.

Lemma 5.3 (An all-weight separating coordinate). *For an integer $w \geq 3$, put $\ell_w(f) = [z^{w-2}aa]\mathcal{O}(\Pi(f))$. For positive integers r, s with $r + s = w$,*

$$\ell_w(z^{r-1}az^{s-1}a) = \frac{\delta_{s,1}}{4}, \quad \ell_w(z^{w-2}a\bar{a}) = -\frac{1}{4}. \quad (92)$$

Moreover ℓ_w vanishes on every word with at most one occurrence of a or $\bar{a}$.

Proof. Write $N = w - 2 > 0$. To contribute to z^Naa , a projected word on $\{z, a, \bar{a}\}$ must retain every zero letter and both nonreal letters. A block consisting only of zero letters cannot do this: its Eulerian image is zero if its length exceeds one, and $\mathcal{R}(z) = -b/2$ otherwise. Also CE of any such block contains no z , since E introduces no c and $\tau(z) = c - b$. Thus only one-block and two-block terms of (88) can contribute, and each block containing a zero letter must use the $E/2$ part of E_+ .

For a block containing just one a , its component with all zeros retained is the usual right-endpoint shuffle regularization

$$z^uaz^v \mapsto (-1)^v \binom{u+v}{v} z^{u+v}a.$$

Indeed setting the shuffle variable z to zero kills every product term in $e(z^uaz^v) - (z^uaz^v)$, because each such product has a factor consisting only of zeros. The displayed formula follows by shuffle regularization, or by induction on v using the shuffle product with z .

For $z^{r-1}az^{s-1}a$, the sum of the two-block shuffle coefficients at z^Naa is therefore

$$\Sigma = 2 \sum_{j=0}^{s-1} (-1)^j \binom{r-1+j}{j} \binom{N}{r-1+j} = 2 \binom{N}{r-1} \sum_{j=0}^{s-1} (-1)^j \binom{s-1}{j} = 2\delta_{s,1}.$$

Convolution exponentiation before the Cayley projection says that the one-block coefficient is $\delta_{s,1} - \Sigma/2$. After projection, the one-block term is multiplied by $1/2$ and the two-block sum by $1/8$. Their total is $\delta_{s,1}/2 - \Sigma/8 = \delta_{s,1}/4$. There is no $z^N\bar{a}\bar{a}$ contribution: obtaining both conjugate letters would require the CE part of both blocks, and at least one block contains a zero letter.

For z^Naa , the only contributing cut is $z^Na \mid \bar{a}$. The first factor contributes $z^Na/2$ and the second contributes $-a/2$ from $C\bar{a} = b - a$. The exponential factor $1/2!$ and the coefficient 2 of z^Naa in $z^Na \sqcup a$ give $-1/4$. Again the conjugate coordinate is absent. Finally, $e, \mathcal{R}, C$ never increase the number of nonreal letters. Every term of (88) consequently has at most one such letter when the original word does, proving the last assertion. $\square$

Proposition 5.4 (Full Cayley-ideal separation). *The rational functional*

$$\ell(f) = [z^5 aa] \mathcal{O}(\Pi(f))$$

annihilates the full ideal $\mathcal{J}$ and satisfies

$$\ell(F) = -166348800 \neq 0. \quad (93)$$

Thus the imaginary lift of the frozen target is not a consequence of convergent Cayley symmetry and its complete shuffle-multiplicative closure.

Proof. Annihilation follows from Theorem 5.2. Lemma 5.3 gives

f	$z^5 aa$	$z^5 a\bar{a}$	$z^3 az^2 a$	$zaz^4 a$	$z^6 a$
$\ell(f)$	$1/4$	$-1/4$	0	0	$0.$

Each word in the three shuffle products in (91) has just one nonreal letter, so those products also have value zero under ℓ . The table gives $(-179712000 - 485683200)/4 = -166348800$. The supplied program replays these exact rational calculations as an independent finite audit of the analytic formula. More explicitly, Π commutes with K , so $\mathcal{O}(\Pi(F)) \neq 0$ implies $\Pi((F - KF)/2) \neq 0$. The kernel theorem therefore gives $(F - KF)/2 \notin \mathcal{J}$, which is the asserted imaginary statement. $\square$

The complete $\mathcal{O}(\Pi(F))$ has 3444 nonzero word coordinates; it is provided as `data/cayley_s6_normal_form.json`. The displayed nonzero coefficient alone is sufficient for separation. This does not contradict the very small certified numerical residual of F . It proves nonmembership in a specified formal ideal whose definition omits double shuffle and other available period identities. In particular, the new certificate does not assert nonmembership in the separate 5131-row presentation, does not enlarge that row presentation by implication, and does not establish or disprove the numerical conjecture.

5.5 Verification and the next reduction step

The independent formulas exposed in the code are useful for auditing the construction. The implementation of e uses the equivalent permutation formula

$$e(v_1 \cdots v_n) = \sum_{\sigma \in S_n} \frac{(-1)^{d(\sigma^{-1})}}{n \binom{n-1}{d(\sigma^{-1})}} v_{\sigma(1)} \cdots v_{\sigma(n)},$$

where d counts descents. It follows directly from (85): a permutation can arise from a cut into k blocks exactly when the $k-1$ cut positions include every descent of the inverse permutation. Summing $(-1)^{k-1} \binom{n-1-d}{k-1-d}/k$ gives the displayed coefficient. The verifier compares this implementation with the consecutive-cut definition on small distinct-letter words. It also checks the projection and product identities on all words and all shuffle products of total weight at most four, then reconstructs the weight-seven target and its separating values with rational arithmetic.

A combined proof search can now replace each additional exact relation R by $\Pi(R)$ and the target by $\Pi(F)$. For any specified family $\mathcal{R} \subset \mathcal{A}$, the elementary equivalence

$$F \in \mathcal{J} + \text{span}_{\mathbb{Q}} \mathcal{R} \iff \Pi(F) \in \text{span}_{\mathbb{Q}} \{\Pi(R) : R \in \mathcal{R}\}$$

follows by subtracting $f - \Pi(f) \in \mathcal{J}$. This supplies an exact, reconstructible way to impose the full Cayley ideal before searching the additional double-shuffle rows. It is a completed reduction tool for that next search, rather than a claim that the remaining search has already succeeded.

6 Manuscript updates and a further research program

The four parts of this article have different mathematical outputs. The Euler theorem proves a uniform inequality and identifies its exact optimal constant. The golden certificates prove special-value and functional identities. The radial theorem classifies a local change in the shape of a zero curve. The Cayley theorem constructs an algebraic quotient and separates a target from a specific relation ideal. Their common feature is that each conclusion has a finite, explicit point of verification, together with an analytic or algebraic argument explaining why that verification suffices.

6.1 Corrections of status and scope

No false displayed formula was established in the source material audited for this continuation. The proposed corrections are therefore specific updates of proof status and precise limitations on how the new results may be used. A blanket assertion that the entire 375-page manuscript has been verified would exceed this audit.

The universal Euler question is settled. The question in `chapters/10-discovery.tex` asking for a universal real-order Euler bound above the critical triangle can be replaced by Theorem 2.1. Its earlier parameter-dependent bounds remain valid and can still be useful when they are smaller. The new result provides the universal budget $9/(8 \cdot 2^N)$ for $N \geq 2$ and identifies the optimal constant for all $N \geq 1$. The exact critical-triangle constant also remains a separate sharper statement on its restricted parameter set.

The source’s warning that the Gaussian maximum alone does not furnish the universal Euler bound is mathematically correct. It should be retained as an explanation of the missing implication, followed by the new positive-tail representation and uniform kernel gap. The proof cannot be shortened by deleting that step. In particular, a bound on the complete sum does not automatically bound every rescaled tail.

The remaining displayed golden weight-five formulas are proved. The numerical-evidence paragraph preceding the second and third weight-five formulas in `chapters/03-algebraic.tex` can now cite Theorem 3.1. The canonical index-12, 20, and 24 normalizations and their lower-weight vanishing statements can be updated using the exact descent and recombination in Section 3. The earlier numerical checks are useful historical evidence but are no longer the justification for these identities.

The weight-6 through weight-9 formulas retain their prior conjectural status. Contracting a weight-five tensor proves identities of lower weight; it cannot create a weight-six tensor identity. Likewise, a successful finite alphabet does not prove that its specializations exhaust all numerical relations at the golden ratio.

The quartic-transition question has a local answer. The decimal transition in `chapters/05-real-turning.tex` can be supplemented by the exact rational rectangle in Theorem 4.3. The higher radial behavior is given by the negative sextic coefficient and Theorem 4.5. There is exactly one transition in the stated rectangle, and an open local region with two small-radius turning points. Neither assertion proves that this is the only transition on the entire threshold curve or that these are the only turning points in the full disk.

The Cayley obstruction is strengthened, while S_6 stays open. The projector in Theorem 5.2 supplies a constructive normal form for the full ideal of convergent Cayley relations and their shuffle

multiples. The exact separating coordinate applies to that whole ideal. It is stronger than a failure to solve one bounded matrix system, but still concerns a specified formal presentation. The conjectural numerical equality for S_6 could be proved by relations that are absent from this ideal. Its S_8 analogue is also untouched. A nonzero formal normal form must not be reported as a nonzero special value.

Two bookkeeping rules should accompany integration. First, rational constant factors in $1 - f$ belong to the golden tensor calculation: the factor 2 in $1 - (1 - t)/(1 + t)$ is a simple example. Removing all constant factors would remove logarithmic information. Second, Cayley-compatible endpoint projection must use the shifted generators $z + b/2$ and $c - b/2$. Setting the unshifted endpoint generators to zero does not commute with the involution. These are requirements of the new proofs, not allegations that the baseline already made either mistake.

The accompanying `integration/INTEGRATION.md` gives concrete source locations, replacement prose, and the label mapping. All integration should be compared with the pinned baseline: later repository changes may already incorporate some of these statements.

6.2 Research questions on uniform bounds and asymptotics

1. Optimize the later Euler steps. For each integer $N \geq 2$, define

$$C_N^{\text{av}} = \sup_{a,b>0} \mathbb{E}[\mathcal{K}_N(X_a, Y_b)], \quad C_N^{\text{pt}} = \max_{0 \leq x, y \leq 1} \mathcal{K}_N(x, y),$$

where the boundary value at $y = 1$ is continuous. The present theorem gives $C_N^{\text{av}} \leq C_N^{\text{pt}} < 9/8$. Determine either sequence sharply. The two optimization problems should be kept separate: the gamma laws form a restricted family of probability measures, so a pointwise maximizer need not be approachable by these averages. A useful first target is an exact algebraic description of the pointwise maximum for $N = 2$, obtained from its rational stationary equations and certified boundary comparisons.

2. Determine the fixed-parameter tail asymptotics. For fixed $a, b > 0$, the exact kernel representation suggests a sharper description than a universal constant times 2^{-N} . The transition of $(1 - x^2 y^2)^N$ occurs near $xy \asymp N^{-1/2}$. The product XY has $-\log(XY)$ gamma distributed with shape $a + b$, so Mellin analysis of that region is natural. The difficulty is to retain the cancellation in $(W_N(xy) - W_N(x))/(1 - y)$ uniformly near $y = 1$. A proof should state its leading term and remainder uniformly on compact positive parameter sets, then study what fails as an order approaches zero. This could give much smaller parameter-sensitive error budgets without weakening the unconditional universal bound.

3. Extend the positive-tail method to other arguments. At the Gaussian point, the alternating series and the rational denominators align particularly well. Which roots of unity, or which fixed angles in the slit plane, admit a transform whose remainder is a positive probability average? The first concrete experiment should be a residue-class decomposition at a primitive sixth root. A useful theorem would identify the angle-dependent kernel and its sign before attempting a global numerical constant. At a general angle, positivity may fail even though the analytic continuation remains well defined.

6.3 Research questions on radial geometry

4. Prove or refute global uniqueness of the quartic transition. The precise unresolved assertion is that $Q(a_*(b), b) = 0$ has only the certified solution for $0 < b < 1$. The determinant calculation already provides a positive derivative along the threshold in a small neighborhood. An effective program would combine analytic endpoint expansions at $b \downarrow 0$ and $b \uparrow 1$ with a finite interval covering of the remaining compact interval. A plot of the threshold is not sufficient, especially near the fully constant point $(1, 1)$, where several coefficients vanish.

5. Make the double-turning neighborhood effective. The local theorem proves the existence of a fixed radius and a parameter neighborhood but does not print their sizes. One can seek explicit bounds for U_{tt} and the fourth and higher Taylor remainders on a rational parameter box and $0 \leq t \leq T$. The all-orders recurrence isolates each coefficient, while a Cauchy bound on the normalized implicit equation can control the infinite tail. The resulting certificate should exhibit actual rational parameter pairs with two certified turning-radius intervals. This is a natural next strengthening of the current existence theorem.

6. Understand the all-orders degeneracy at $(1, 1)$. At that point $F_{1,1}(z) = \frac{1}{2} \log^2(1 - z)$ and the normalized radius is identically $1/2$. The nearby double-turning transition has a nonzero sextic coefficient. Determine the first-order parameter variation of the entire function Φ , rather than one Taylor coefficient at a time, and ask whether it constrains the number of small-radius oscillations. This may clarify the relationship between the isolated certified transition and the infinitely degenerate constant curve. It must also be reconciled with the still-open full-radius motion conjectures for integer outer order.

6.4 Research questions on identities and relation spaces

7. Lift the golden weight-six and weight-seven candidates. The two higher reconstructions in the manuscript are concrete next targets. Search for a rational-function tensor certificate of the correct weight whose specialization is exactly the proposed vector, including its logarithmic completion. Gangl's work [6] gives relevant higher-weight functional identities and uses richer multiplicative alphabets. A search restricted to $t, 1 - t, 1 + t$ should therefore be treated as one finite model. The essential deliverables are an exact tensor row, an endpoint constant, and ordinary-polylogarithm reconstruction, not only a nullspace rank. The weight-eight and weight-nine formulas require a further step and should not be assumed to follow once weights six and seven are solved.

8. Simplify and classify the functional certificates. Can the 58- and 60-term rows be expressed as short combinations of classical Kummer specializations? Such a decomposition could reveal why the golden specialization eliminates the unwanted powers. A separate finite problem is to determine the exact image of the full degree-bounded tensor kernel under golden specialization. Its rank and nullspace can be certified by rational linear algebra. Any minimality result must explicitly retain the alphabet and degree bound; it would not establish independence of the resulting periods.

9. Specialize the new functional identities in other fields. The identities (46) hold for all $0 < t < 1$, so their use is not confined to the golden ratio. For example, with $\sigma = \sqrt{2} - 1$, one has $1 - \sigma = \sqrt{2}\sigma$ and $1 + \sigma = \sqrt{2}$. Every table argument then becomes

$$_s \sigma^{a+b} 2^{(b+c)/2}.$$

Substitution gives two exact weight-five relations in $\mathbb{Q}(\sqrt{2})$ with the same displayed endpoint constants. The research problem is to combine them with duplication, inversion, and other field relations until a small useful basis emerges. This provides a structured source of new candidate evaluations. It avoids guessing a large unrelated list of numerical constants at the outset.

10. Add the missing relations to the Cayley quotient. For a proposed collection of additional relation vectors $\mathcal{T}$, the exact next computation is to ask whether

$$\Pi(F) \in \text{span}_{\mathbb{Q}}\{\Pi(T) : T \in \mathcal{T}\}.$$

The equivalence with membership in the original Cayley ideal plus these additional rows follows from the kernel theorem. Thus every candidate stuffle, distribution, or complementary-argument identity can be projected immediately. Start with convergent relations and lower-weight products, then introduce regularized relations only with their analytic conventions stated explicitly. The nonzero coordinate -166348800 identifies information that the added relations must actually supply. If the enlarged presentation still fails, report that precise obstruction; it would not settle numerical independence.

11. Make the quotient computationally practical in higher weight. The explicit word projector is finite, but its expanded support grows quickly. A representation by Cayley-positive Eulerian generators can retain products in factored form and postpone full shuffle expansion. Determine the cost of converting a specified target to these coordinates, and compare it with elimination in a relation matrix. A useful implementation should return a reconstruction certificate, so that compression never hides the formal identity it represents. The all-weight separating-coordinate lemma suggests that many target tests may need only a few coordinates of the projector.

12. Formalize the smallest complete proof path. The Euler result offers a manageable entry point: polynomial identities, rational Bernstein positivity, the probability-kernel formula, and the truncated-power mixture. The golden result offers a different path through finite tensor identities and one-variable calculus. A proof-assistant project should first choose one complete analytic theorem, rather than merely formalizing its arithmetic receipt. In either case, the present records separate the finite equalities from the mathematical implications they support.

6.5 Reproduction and the boundary of the deliverable

The archive is arranged so that the proofs and their finite premises can be inspected independently. The article contains all definitions, the golden coefficient tables, the three Euler polynomials, and the local radial coefficient formulas. The data files contain the full Bernstein partitions, the rational interval bounds, the exact golden rows and their specialization receipts, and the Cayley normal form. The independent review notes explain the delicate analytic and algebraic steps.

From the extracted directory, the commands

```
python3 -m pip install -r requirements.txt
python3 code/run_all.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

replay the certificates and rebuild the article. The requirements file lists the Python dependencies used for replay and figure generation; the two core Euler and radial producers use only Python's standard library. The golden verifier and independent Euler reviewer use SymPy for exact symbolic

arithmetic. The Cayley verifier uses rational arithmetic throughout. Figure generation is optional and is invoked by adding `-figures` to the replay command.

No integer-relation search is needed to verify the accepted golden identities, and no numerical root solver is needed to certify the radial rectangle. The figures visualize the proved structure and are explicitly labeled as diagnostic or as a limiting local model. The open questions above remain open in this deliverable; they are not silently promoted by successful numerical comparisons or by the strength of a neighboring theorem.

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